Operational Data Introduction A163806

Information in this chapter includes operations that are peculiar to the operation of the Boeing 737. This information has been evaluated and/or demonstrated for Type Certification and operational use.

Operation in Icing Conditions - Engine Anti-Ice System A163806

Turn engine ignition on prior to activating the engine anti-ice system. Turn ignition off after the engine anti-ice is turned off.

For engine anti-ice operation see Engine Appendix, 'Engines - Anti-Ice System'.

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

Page Intentionally Left Blank

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

Performance Introduction A163806

The Performance in the basic Airplane Flight Manual are applicable except as amended herein.

Performance Software Serial Number (PSSN) A191108

D631A002.E08.G010

Airplane Flight Manual - Digital Performance Information (AFM-DPI) A191108

The following is a summary description of the AFM-DPI software denoted by the Performance Software Serial Number (PSSN) below. Refer to the Airplane Serial Number and Appendix Effectivity Table of the basic manual for a complete listing of the airplane serial numbers authorized to use this Performance appendix.

To access and run the AFM-DPI software and data shown below, install the PSSN into the Boeing Performance Engineers Tool (PET) software version shown below. Instructions for installing the PET software and importing the AFM-DPI data into PET are contained in the AFM-DPI Users Guide.

Model	Certification Basis	Brake Category	Engine Thrust
737-8	EASA	B	LEAP-1B27

D631A002.E08.G010

Software Identification			
Performance Software Serial Number	Software Version	Data Version	PET Version
D631A002.E08.G010	16.00	S003	3.0 or greater

Airplane Flight Manual - Digital Performance Information (AFM-DPI) Release A163806

Release notes for the software and data are available from the PET interface. Select the "Certified Performance (AFM-DPI)" option within PET. The release notes for all versions of AFM-DPI that have been imported into PET will be available under the "PSSN Versions" button on the input screen.

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

Airplane Flight Manual Performance Adjustment - Alternate EEC Mode A170351

The following is applicable only to EEC software version 4.5 or earlier.

When operating in the alternate EEC mode, additional performance adjustments should be applied.

1. Reduce all thrust rating limits (takeoff, go around, max continuous thrust) by 2% N1.

2. The performance limited weights are reduced by the amounts shown in the table below:

Takeoff	Enroute Climb	Approach & Landing
14,000 lbs (6,350 kg)	17,000 lbs (7,711 kg)	8,700 lbs (3,946 kg)

3. Do not use Autothrottle during Takeoff, Climb or Max Continuous Thrust operation (Autothrottle will chase the FMC target, which will prevent the reduction in rating limits above).

Alternate Performance Capability A191201

Alternate EEC Mode
Ten Minute Takeoff Thrust
Gear Down Dispatch
Forward CG Limit
Forward CG Limit 2* - Takeoff and Enroute
Alternate Forward CG - Takeoff and Landing
Alternate Forward CG 2* - Takeoff
Two Position Tailskid
Alternate Flaps 5/30 Go-Around
Alternate Flaps 5/30 Go-Around including Autoland

* Forward CG Limit 2 (Alternate Forward CG 2, if applicable) must be selected in AFM-DPI when specified in the Weight and Center of Gravity Limitations Section of the basic manual. If not specified, then Forward CG Limit (Alternate Forward CG, if applicable) must be selected.

Note:
Specific airplane serial numbers must be operated in accordance with the Authorized Alternate Performance Capability specified in the basic Airplane Flight Manual.

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

Engine Power Setting A170428

TAKEOFF POWER SETTING

Takeoff settings can be determined by use of the approved AFM-DPI software application.

GO-AROUND POWER SETTING

The Go-Around Power Setting can be determined by use of the approved AFM-DPI software application.

INFLIGHT POWER SETTING

The Inflight Power Setting for maximum continuous power rating for determining inflight N1 power setting is provided in the approved AFM-DPI software application.

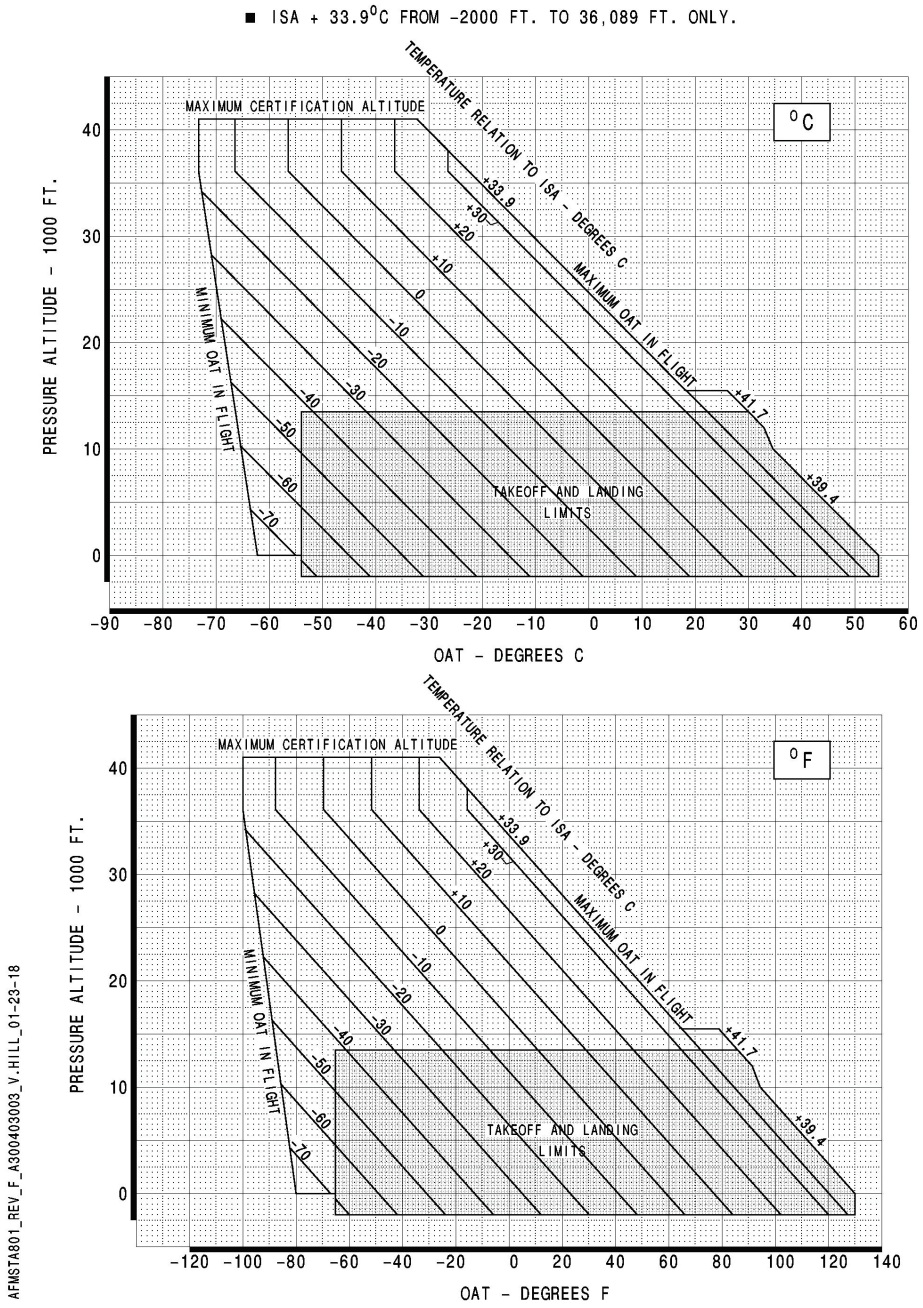
AFM-DPI SELECTION OF ANTI-ICE

To ensure that the appropriate anti-ice corrections are applied to engine power settings when using AFM-DPI, refer to the following table. The correct selection of anti-ice in AFM-DPI is determined from the engine and wing anti-ice configuration as shown below.

AFM-DPI SELECTION OF ANTI-ICE		
ENGINE ANTI-ICE	WING ANTI-ICE	AFM-DPI SELECTION
OFF	OFF	OFF
ON	OFF	ENGINE
ON	ON	ENG & WING

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

Environmental Envelope A180043

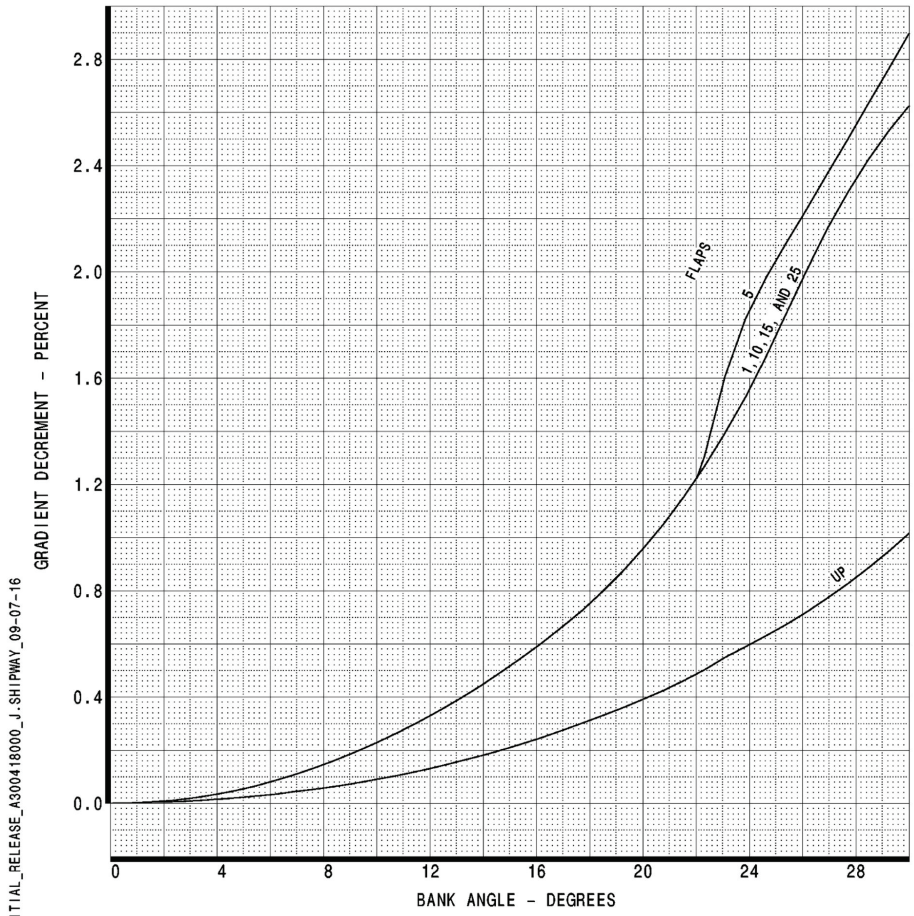


Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

Gradient Reduction for Banked Flight A163806

Corrections to the flight path gradient are shown as Gradient Decrement for steady turns in the flight path.

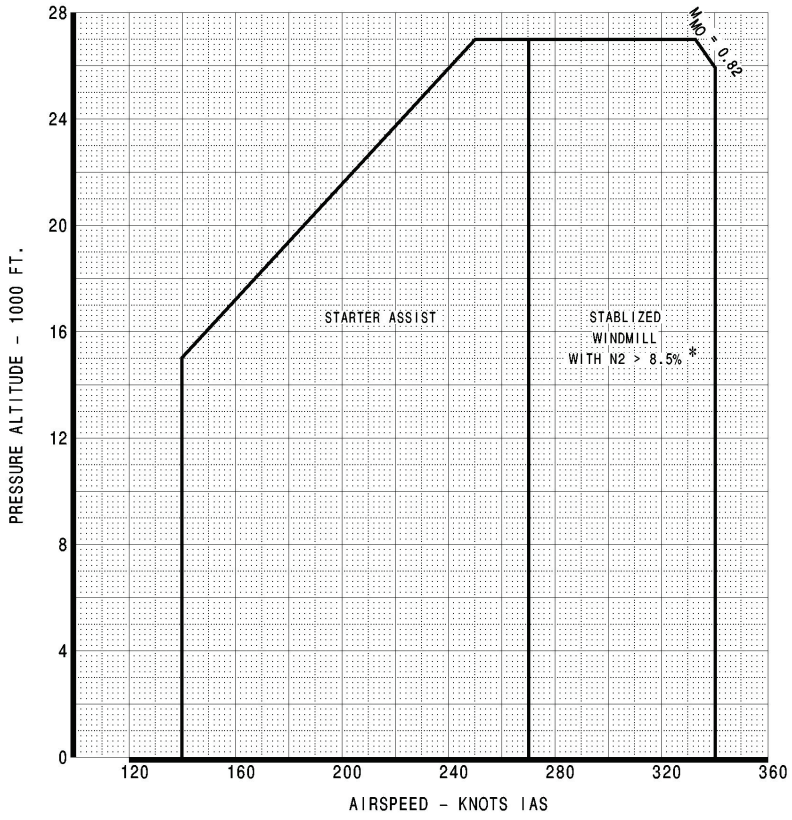
- TAKEOFF FLAP DETENTS ARE EVALUATED AT V_2.
- FLAPS UP IS EVALUATED AT THE FLAPS UP MANEUVER SPEED.



Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

Inflight Restart Envelope A163806

- SUCCESSFUL STARTS ARE NOT ASSURED OUTSIDE OF THIS ENVELOPE.
- WINDMILL START PROCEDURE ENVELOPE APPLIES TO STABILIZED WINDMILL.
- AT AIRSPEEDS WITHIN THE AMBERBAND, STARTER ASSIST RESTARTS MAY BE DELAYED.



*HYDRAULIC ENGINE DRIVEN PUMP MUST BE MANUALLY DEPRESSURIZED BEFORE STABILIZED WINDMILLING RESTART

AFMSTA801_INITIAL_RELEASE_A300417000_S.HARIRAM_09-13-16

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

Noise - Characteristics A170387

Care should be exercised in selecting the page that corresponds with the specific maximum engine thrust of the current airplane. The specific airplane configuration applicability is provided on each version of this appendix.

CERTIFICATED NOISE LEVELS

The following noise levels were established using test data obtained and analyzed under procedures of 14 CFR Part 36 and ICAO Annex 16. This aircraft complies with 14 CFR Part 36, Stage 4 and ICAO Annex 16, Chapter 4 noise level requirements.

NOISE CHARACTERISTICS APPLICABILITY

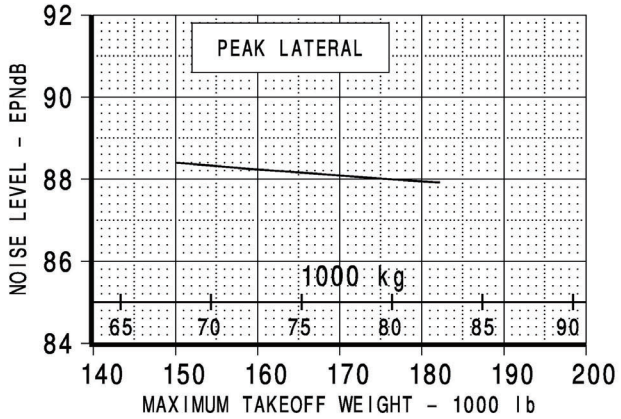
No determination has been made by the Federal Aviation Administration or the European Aviation Safety Agency that the noise levels of this aircraft are, or should be, acceptable or unacceptable for operation at, into, or out of, any airport.

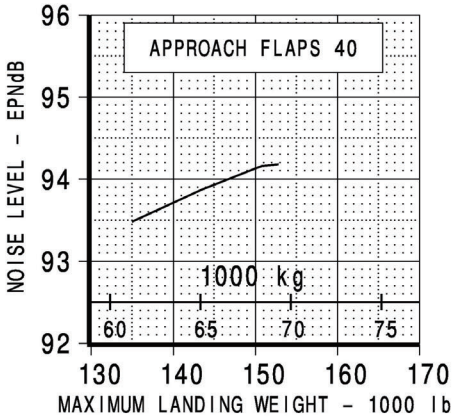
Noise levels are determined by entering the graphs at the maximum weights defined under Certificate Limitations in the Airplane Flight Manual.

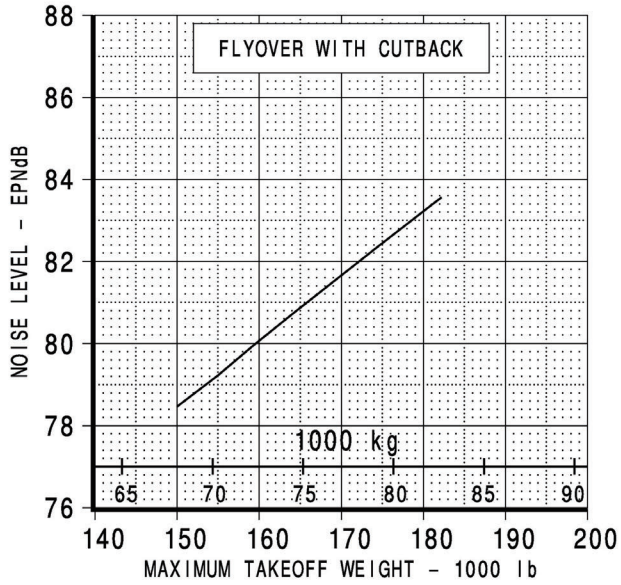
Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

Noise - Configuration A191108

Engines: CFM LEAP-1B27
Model: 737-8 with One Position Tailskid
Nacelle Treatment: Basic Configuration Including Inlet and Fan Duct Treatments







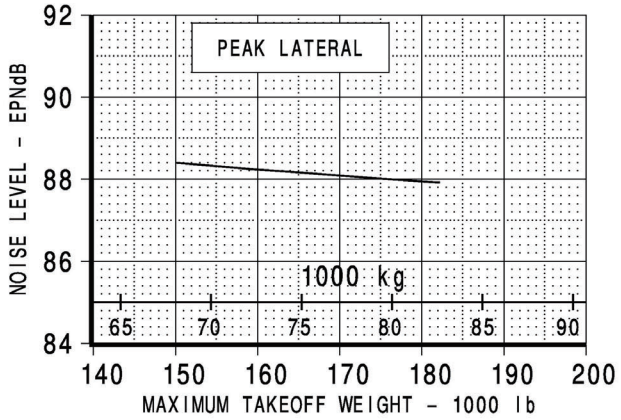
Noise Configuration Code:
3M8.R4.LEAP1B27.BLK2.1PTS

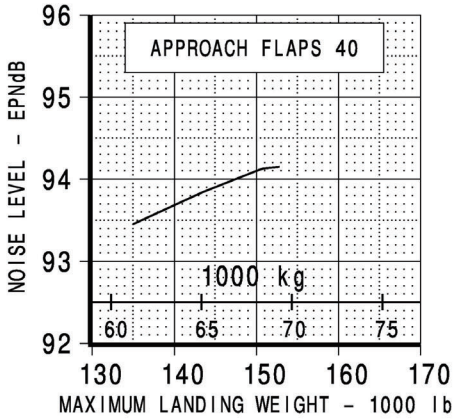
Noise - Configuration A191108

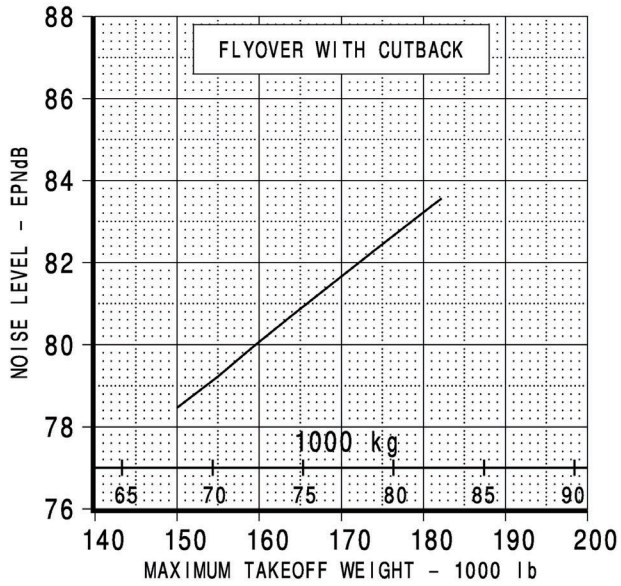
Engines: CFM LEAP-1B27
Model: 737-8 with Two Position Tailskid
Nacelle Treatment: Basic Configuration Including Inlet and Fan Duct Treatments

Continued on next page

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.







Noise Configuration Code:
3M8.R4.LEAP1B27.BLK2.2PTS

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

Non-Normal Configuration Landing Distances A170245

737-8 Non-Normal Configuration Landing Distances - Dry Runway - 1 Position Tailskid
LEAP-1B Engines

LANDING DISTANCES AND ADJUSTMENTS (FEET)								
QRH NON-NORMAL CHECKLIST	REFERENCE SPEED (VREF)	REFERENCE DISTANCE FOR 150000 LBS	WEIGHT ADJUSTMENT PER 10000 POUNDS		ALTITUDE ADJUSTMENT PER 1000 FT ABOVE SEA LEVEL *		RUNWAY WIND ADJUSTMENT PER 10 KNOTS	
			ABOVE 150000 LBS	BELOW 150000 LBS	STD	HIGH	HEAD WIND	TAIL WIND
Airspeed Unreliable (Flaps 15)	VREF15	5340	250	-270	140	170	-210	690
Airspeed Unreliable (Flaps 30)	VREF30	5000	230	-260	130	160	-210	670
Airspeed Unreliable (Flaps 40)	VREF40	4670	260	-230	130	160	-200	650
All Flaps Up Landing (Flaps UP)	VREF40+55	6950	370	-290	190	240	-240	780
ANTISKID INOPERATIVE (Flaps 15)	VREF15	6650	360	-380	200	250	-330	1130
ANTISKID INOPERATIVE (Flaps 30)	VREF30	6120	320	-350	180	230	-320	1090
ANTISKID INOPERATIVE (Flaps 40)	VREF40	5750	360	-320	170	220	-310	1070
Jammed or Restricted Flight Controls (Flaps 15)	VREF15+10	5340	250	-270	140	170	-210	690
LEADING EDGE FLAPS (Flaps 15)	VREF15+15	5590	250	-280	140	180	-220	710
LOSS OF BOTH ENGINE DRIVEN GENERATORS (Flaps 15)	VREF15	7830	450	-470	250	330	-370	1260
LOSS OF BOTH ENGINE DRIVEN GENERATORS (Flaps 30)	VREF30	7320	410	-440	230	300	-360	1230
LOSS OF BOTH ENGINE DRIVEN GENERATORS (Flaps 40)	VREF40	6890	480	-410	230	290	-360	1210
LOSS OF SYSTEM A (Flaps 15)	VREF15	4960	260	-270	130	160	-210	670
LOSS OF SYSTEM A (Flaps 30)	VREF30	4600	240	-250	120	150	-200	650
LOSS OF SYSTEM A (Flaps 40)	VREF40	4290	280	-230	120	150	-200	630
LOSS OF SYSTEM A AND SYSTEM B (Flaps 15)	VREF15	6340	350	-340	170	220	-270	880
LOSS OF SYSTEM B (Flaps 15)	VREF15	5010	270	-270	130	170	-220	730
MANUAL REVERSION (Flaps 15)	VREF15	6340	350	-340	170	220	-270	880

Actual (unfactored) distances are shown.

Includes a distance from runway threshold to touchdown associated with a flare time of 7 seconds.

Assumes maximum autobrake setting and maximum available reverse thrust.

Manual braking is assumed for ANTISKID INOPERATIVE, LOSS OF BOTH ENGINE DRIVEN GENERATORS, LOSS OF SYSTEM B, LOSS OF SYSTEM A AND SYSTEM B and MANUAL REVERSION since autobrakes are inoperative.

*For landing distance at or below 8000 ft pressure altitude, apply the STD altitude adjustment.

For altitudes higher than 8000 ft, first apply the STD altitude adjustment to 8000 ft, then apply the HIGH altitude adjustment for the incremental altitude above 8000 ft.

Continued on next page

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

737-8 Non-Normal Configuration Landing Distances - Dry Runway - 1 Position Tailskid
LEAP-1B Engines

LANDING DISTANCES AND ADJUSTMENTS (FEET)								
QRH NON-NORMAL CHECKLIST	REFERENCE SPEED (VREF)	RUNWAY SLOPE ADJUSTMENT PER 1%		TEMPERATURE ADJUSTMENT PER		APPROACH SPEED ADJUSTMENT PER 5 KNOTS ABOVE V_{REF}	REVERSE THRUST ADJUSTMENT	
		DOWN HILL	UP HILL	10° C ABOVE ISA	10° C BELOW ISA		ONE REVERSER	NO REVERSER
Airspeed Unreliable (Flaps 15)	VREF15	10	0	140	-140	N/A	10	30
Airspeed Unreliable (Flaps 30)	VREF30	10	0	130	-130	N/A	10	30
Airspeed Unreliable (Flaps 40)	VREF40	20	-10	120	-120	N/A	10	30
All Flaps Up Landing (Flaps UP)	VREF40+55	10	0	190	-190	280	10	40
ANTISKID INOPERATIVE (Flaps 15)	VREF15	200	-160	180	-180	270	510	1140
ANTISKID INOPERATIVE (Flaps 30)	VREF30	180	-150	170	-170	250	410	900
ANTISKID INOPERATIVE (Flaps 40)	VREF40	170	-140	160	-160	250	360	780
Jammed or Restricted Flight Controls (Flaps 15)	VREF15+10	10	0	140	-140	250	10	30
LEADING EDGE FLAPS (Flaps 15)	VREF15+15	20	-10	150	-150	270	10	40
LOSS OF BOTH ENGINE DRIVEN GENERATORS (Flaps 15)	VREF15	300	-240	220	-220	380	920	2150
LOSS OF BOTH ENGINE DRIVEN GENERATORS (Flaps 30)	VREF30	280	-230	200	-210	370	780	1790
LOSS OF BOTH ENGINE DRIVEN GENERATORS (Flaps 40)	VREF40	280	-220	190	-200	370	690	1560
LOSS OF SYSTEM A (Flaps 15)	VREF15	40	-20	130	-130	260	80	140
LOSS OF SYSTEM A (Flaps 30)	VREF30	30	-20	120	-120	240	60	110
LOSS OF SYSTEM A (Flaps 40)	VREF40	40	-20	110	-110	240	60	100
LOSS OF SYSTEM A AND SYSTEM B (Flaps 15)	VREF15	160	-140	170	-170	370	60	310
LOSS OF SYSTEM B (Flaps 15)	VREF15	80	-70	130	-130	220	140	240
MANUAL REVERSION (Flaps 15)	VREF15	160	-140	170	-170	370	60	310

Actual (unfactored) distances are shown.

Includes a distance from runway threshold to touchdown associated with a flare time of 7 seconds.

Assumes maximum autobrake setting and maximum available reverse thrust.

Manual braking is assumed for ANTISKID INOPERATIVE, LOSS OF BOTH ENGINE DRIVEN GENERATORS, LOSS OF SYSTEM B, LOSS OF SYSTEM A AND SYSTEM B and MANUAL REVERSION since autobrakes are inoperative.

Continued on next page

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

737-8 Non-Normal Configuration Landing Distances - Dry Runway - 1 Position Tailskid
LEAP-1B Engines

LANDING DISTANCES AND ADJUSTMENTS (FEET)								
QRH NON-NORMAL CHECKLIST	REFERENCE SPEED (VREF)	REFERENCE DISTANCE FOR 150000 LBS	WEIGHT ADJUSTMENT PER 10000 POUNDS		ALTITUDE ADJUSTMENT PER 1000 FT ABOVE SEA LEVEL *		RUNWAY WIND ADJUSTMENT PER 10 KNOTS	
			ABOVE 150000 LBS	BELOW 150000 LBS	STD	HIGH	HEAD WIND	TAIL WIND
One Engine Inoperative Landing (Flaps 15)	VREF15	4860	240	-260	120	150	-210	670
One Engine Inoperative Landing (Flaps 30)	VREF30	4530	220	-240	110	140	-200	650
SPOILERS (Flaps 15)	VREF15	5030	270	-280	140	170	-210	680
SPOILERS (Flaps 30)	VREF30	4750	250	-270	130	160	-210	670
SPOILERS (Flaps 40)	VREF40	4470	290	-250	130	170	-200	660
Stabilizer Trim Inoperative (Flaps 15)	VREF15+10	5340	250	-270	140	170	-210	690
Trailing Edge Flap Asymmetry (1 =< Flap Lever =< 15)	VREF40+30	5630	290	-260	150	190	-220	710
Trailing Edge Flap Asymmetry (Flap Lever 15 or 25)	VREF15	4860	230	-260	120	150	-210	670
Trailing Edge Flap Asymmetry (Flap Lever 30)	VREF30	4530	220	-240	110	140	-200	650
Trailing Edge Flap Disagree (1 =< Indicated Flap < 15)	VREF40+30	5630	290	-260	150	190	-220	710
Trailing Edge Flap Disagree (15 =< Indicated Flap < 30)	VREF15	4860	230	-260	120	150	-210	670
Trailing Edge Flap Disagree (30 =< Indicated Flap < 40)	VREF30	4530	220	-240	110	140	-200	650
Trailing Edge Flaps Up Landing (Flaps Up)	VREF40+40	6140	320	-270	170	210	-230	740

Actual (unfactored) distances are shown.

Includes a distance from runway threshold to touchdown associated with a flare time of 7 seconds.

Assumes maximum autobrake setting and maximum available reverse thrust.

For "One Engine Inoperative Landing - Flaps 30" autoland, increase the landing distance by 750 feet.

*For landing distance at or below 8000 ft pressure altitude, apply the STD altitude adjustment.

For altitudes higher than 8000 ft, first apply the STD altitude adjustment to 8000 ft,

then apply the HIGH altitude adjustment for the incremental altitude above 8000 ft.

Continued on next page

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

737-8 Non-Normal Configuration Landing Distances - Dry Runway - 1 Position Tailskid
LEAP-1B Engines

LANDING DISTANCES AND ADJUSTMENTS (FEET)								
QRH NON-NORMAL CHECKLIST	REFERENCE SPEED (VREF)	RUNWAY SLOPE ADJUSTMENT PER 1%		TEMPERATURE ADJUSTMENT PER		APPROACH SPEED ADJUSTMENT PER 5 KNOTS ABOVE V_{REF}	REVERSE THRUST ADJUSTMENT	
		DOWN HILL	UP HILL	10° C ABOVE ISA	10° C BELOW ISA		ONE REVERSER	NO REVERSER
One Engine Inoperative Landing (Flaps 15)	VREF15	10	0	130	-130	240	N/A	20
One Engine Inoperative Landing (Flaps 30)	VREF30	10	0	120	-120	230	N/A	10
SPOILERS (Flaps 15)	VREF15	50	-30	130	-130	280	180	290
SPOILERS (Flaps 30)	VREF30	60	-40	130	-130	280	160	260
SPOILERS (Flaps 40)	VREF40	60	-50	120	-120	290	140	230
Stabilizer Trim Inoperative (Flaps 15)	VREF15+10	10	0	140	-140	250	10	30
Trailing Edge Flap Asymmetry (1 =< Flap Lever =< 15)	VREF40+30	10	0	150	-150	260	10	30
Trailing Edge Flap Asymmetry (Flap Lever 15 or 25)	VREF15	10	0	130	-130	240	10	30
Trailing Edge Flap Asymmetry (Flap Lever 30)	VREF30	10	0	120	-120	230	10	30
Trailing Edge Flap Disagree (1 =< Indicated Flap < 15)	VREF40+30	10	0	150	-150	260	10	30
Trailing Edge Flap Disagree (15 =< Indicated Flap < 30)	VREF15	10	0	130	-130	240	10	30
Trailing Edge Flap Disagree (30 =< Indicated Flap < 40)	VREF30	10	0	120	-120	230	10	30
Trailing Edge Flaps Up Landing (Flaps Up)	VREF40+40	10	0	160	-170	270	10	30

Actual (unfactored) distances are shown.

Includes a distance from runway threshold to touchdown associated with a flare time of 7 seconds.

Assumes maximum autobrake setting and maximum available reverse thrust.

For "One Engine Inoperative Landing - Flaps 30" autoland, increase the landing distance by 750 feet.

Continued on next page

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

737-8 Non-Normal Configuration Landing Distances - Wet Runway - 1 Position Tailskid
LEAP-1B Engines

LANDING DISTANCES AND ADJUSTMENTS (FEET)								
QRH NON-NORMAL CHECKLIST	REFERENCE SPEED (VREF)	REFERENCE DISTANCE FOR 150000 LBS	WEIGHT ADJUSTMENT PER 10000 POUNDS		ALTITUDE ADJUSTMENT PER 1000 FT ABOVE SEA LEVEL *		RUNWAY WIND ADJUSTMENT PER 10 KNOTS	
			ABOVE 150000 LBS	BELOW 150000 LBS	STD	HIGH	HEAD WIND	TAIL WIND
Airspeed Unreliable (Flaps 15)	VREF15	6110	370	-360	210	280	-300	1040
Airspeed Unreliable (Flaps 30)	VREF30	5620	330	-330	190	250	-280	990
Airspeed Unreliable (Flaps 40)	VREF40	5270	370	-300	180	250	-270	960
All Flaps Up Landing (Flaps UP)	VREF40+55	7990	480	-420	300	400	-350	1200
ANTISKID INOPERATIVE (Flaps 15)	VREF15	7800	520	-500	280	390	-430	1590
ANTISKID INOPERATIVE (Flaps 30)	VREF30	7070	460	-450	250	340	-410	1510
ANTISKID INOPERATIVE (Flaps 40)	VREF40	6570	490	-410	240	320	-400	1460
Jammed or Restricted Flight Controls (Flaps 15)	VREF15+10	6110	370	-360	210	280	-300	1040
LEADING EDGE FLAPS (Flaps 15)	VREF15+15	6530	390	-380	220	310	-310	1080
LOSS OF BOTH ENGINE DRIVEN GENERATORS (Flaps 15)	VREF15	9260	670	-620	370	510	-500	1790
LOSS OF BOTH ENGINE DRIVEN GENERATORS (Flaps 30)	VREF30	8500	590	-560	320	450	-470	1720
LOSS OF BOTH ENGINE DRIVEN GENERATORS (Flaps 40)	VREF40	7910	650	-520	310	430	-460	1670
LOSS OF SYSTEM A (Flaps 15)	VREF15	6330	430	-410	230	330	-330	1170
LOSS OF SYSTEM A (Flaps 30)	VREF30	5680	370	-360	200	280	-310	1090
LOSS OF SYSTEM A (Flaps 40)	VREF40	5220	420	-330	200	270	-290	1040
LOSS OF SYSTEM A AND SYSTEM B (Flaps 15)	VREF15	8600	570	-550	320	440	-440	1530
LOSS OF SYSTEM B (Flaps 15)	VREF15	6380	420	-400	230	310	-340	1190
MANUAL REVERSION (Flaps 15)	VREF15	8600	570	-550	320	440	-440	1530

Actual (unfactored) distances are shown.

Includes a distance from runway threshold to touchdown associated with a flare time of 7 seconds.

Assumes maximum autobrake setting and maximum available reverse thrust.

Manual braking is assumed for ANTISKID INOPERATIVE, LOSS OF BOTH ENGINE DRIVEN GENERATORS, LOSS OF SYSTEM B, LOSS OF SYSTEM A AND SYSTEM B and MANUAL REVERSION since autobrakes are inoperative.

*For landing distance at or below 8000 ft pressure altitude, apply the STD altitude adjustment.

For altitudes higher than 8000 ft, first apply the STD altitude adjustment to 8000 ft, then apply the HIGH altitude adjustment for the incremental altitude above 8000 ft.

Continued on next page

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

737-8 Non-Normal Configuration Landing Distances - Wet Runway - 1 Position Tailskid
LEAP-1B Engines

LANDING DISTANCES AND ADJUSTMENTS (FEET)								
QRH NON-NORMAL CHECKLIST	REFERENCE SPEED (VREF)	RUNWAY SLOPE ADJUSTMENT PER 1%		TEMPERATURE ADJUSTMENT PER		APPROACH SPEED ADJUSTMENT PER 5 KNOTS ABOVE V_{REF}	REVERSE THRUST ADJUSTMENT	
		DOWN HILL	UP HILL	10° C ABOVE ISA	10° C BELOW ISA		ONE REVERSER	NO REVERSER
Airspeed Unreliable (Flaps 15)	VREF15	120	-100	190	-190	N/A	360	730
Airspeed Unreliable (Flaps 30)	VREF30	100	-90	180	-170	N/A	270	540
Airspeed Unreliable (Flaps 40)	VREF40	100	-80	160	-160	N/A	230	450
All Flaps Up Landing (Flaps UP)	VREF40+55	160	-140	260	-260	320	640	1330
ANTISKID INOPERATIVE (Flaps 15)	VREF15	310	-250	250	-250	330	980	2510
ANTISKID INOPERATIVE (Flaps 30)	VREF30	270	-220	220	-220	310	740	1830
ANTISKID INOPERATIVE (Flaps 40)	VREF40	260	-210	200	-200	310	630	1510
Jammed or Restricted Flight Controls (Flaps 15)	VREF15+10	120	-100	190	-190	300	360	730
LEADING EDGE FLAPS (Flaps 15)	VREF15+15	130	-110	210	-210	310	430	880
LOSS OF BOTH ENGINE DRIVEN GENERATORS (Flaps 15)	VREF15	480	-370	300	-300	450	1790	5190
LOSS OF BOTH ENGINE DRIVEN GENERATORS (Flaps 30)	VREF30	440	-340	270	-270	420	1420	3880
LOSS OF BOTH ENGINE DRIVEN GENERATORS (Flaps 40)	VREF40	420	-320	250	-250	420	1200	3170
LOSS OF SYSTEM A (Flaps 15)	VREF15	200	-170	210	-210	380	510	960
LOSS OF SYSTEM A (Flaps 30)	VREF30	170	-140	190	-190	350	360	640
LOSS OF SYSTEM A (Flaps 40)	VREF40	150	-130	170	-170	340	280	490
LOSS OF SYSTEM A AND SYSTEM B (Flaps 15)	VREF15	390	-310	280	-270	510	460	1940
LOSS OF SYSTEM B (Flaps 15)	VREF15	170	-150	210	-210	320	510	980
MANUAL REVERSION (Flaps 15)	VREF15	390	-310	280	-270	510	460	1940

Actual (unfactored) distances are shown.

Includes a distance from runway threshold to touchdown associated with a flare time of 7 seconds.

Assumes maximum autobrake setting and maximum available reverse thrust.

Manual braking is assumed for ANTISKID INOPERATIVE, LOSS OF BOTH ENGINE DRIVEN GENERATORS, LOSS OF SYSTEM B, LOSS OF SYSTEM A AND SYSTEM B and MANUAL REVERSION since autobrakes are inoperative.

Continued on next page

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

737-8 Non-Normal Configuration Landing Distances - Wet Runway - 1 Position Tailskid
LEAP-1B Engines

LANDING DISTANCES AND ADJUSTMENTS (FEET)								
QRH NON-NORMAL CHECKLIST	REFERENCE SPEED (VREF)	REFERENCE DISTANCE FOR 150000 LBS	WEIGHT ADJUSTMENT PER 10000 POUNDS		ALTITUDE ADJUSTMENT PER 1000 FT ABOVE SEA LEVEL *		RUNWAY WIND ADJUSTMENT PER 10 KNOTS	
			ABOVE 150000 LBS	BELOW 150000 LBS	STD	HIGH	HEAD WIND	TAIL WIND
One Engine Inoperative Landing (Flaps 15)	VREF15	5720	370	-360	200	270	-300	1070
One Engine Inoperative Landing (Flaps 30)	VREF30	5210	330	-320	170	240	-280	1000
SPOILERS (Flaps 15)	VREF15	6590	470	-440	250	360	-350	1210
SPOILERS (Flaps 30)	VREF30	6120	420	-410	230	320	-330	1150
SPOILERS (Flaps 40)	VREF40	5700	490	-370	230	320	-320	1110
Stabilizer Trim Inoperative (Flaps 15)	VREF15+10	6110	370	-360	210	280	-300	1040
Trailing Edge Flap Asymmetry (1 =< Flap Lever =< 15)	VREF40+30	6350	410	-350	230	310	-300	1060
Trailing Edge Flap Asymmetry (Flap Lever 15 or 25)	VREF15	5530	350	-340	190	260	-280	990
Trailing Edge Flap Asymmetry (Flap Lever 30)	VREF30	5080	310	-310	170	230	-260	940
Trailing Edge Flap Disagree (1 =< Indicated Flap < 15)	VREF40+30	6350	410	-350	230	310	-300	1060
Trailing Edge Flap Disagree (15 =< Indicated Flap < 30)	VREF15	5530	350	-340	190	260	-280	990
Trailing Edge Flap Disagree (30 =< Indicated Flap < 40)	VREF30	5080	310	-310	170	230	-260	940
Trailing Edge Flaps Up Landing (Flaps Up)	VREF40+40	6910	430	-370	250	340	-320	1110

Actual (unfactored) distances are shown.

Includes a distance from runway threshold to touchdown associated with a flare time of 7 seconds.

Assumes maximum autobrake setting and maximum available reverse thrust.

For "One Engine Inoperative Landing - Flaps 30" autoland, increase the landing distance by 750 feet.

*For landing distance at or below 8000 ft pressure altitude, apply the STD altitude adjustment.

For altitudes higher than 8000 ft, first apply the STD altitude adjustment to 8000 ft,

then apply the HIGH altitude adjustment for the incremental altitude above 8000 ft.

Continued on next page

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

737-8 Non-Normal Configuration Landing Distances - Wet Runway - 1 Position Tailskid
LEAP-1B Engines

LANDING DISTANCES AND ADJUSTMENTS (FEET)								
QRH NON-NORMAL CHECKLIST	REFERENCE SPEED (VREF)	RUNWAY SLOPE ADJUSTMENT PER 1%		TEMPERATURE ADJUSTMENT PER		APPROACH SPEED ADJUSTMENT PER 5 KNOTS ABOVE V_{REF}	REVERSE THRUST ADJUSTMENT	
		DOWN HILL	UP HILL	10° C ABOVE ISA	10° C BELOW ISA		ONE REVERSER	NO REVERSER
One Engine Inoperative Landing (Flaps 15)	VREF15	120	-100	190	-190	320	N/A	320
One Engine Inoperative Landing (Flaps 30)	VREF30	100	-90	170	-170	290	N/A	230
SPOILERS (Flaps 15)	VREF15	240	-200	230	-230	430	680	1500
SPOILERS (Flaps 30)	VREF30	220	-180	210	-210	420	560	1190
SPOILERS (Flaps 40)	VREF40	210	-170	190	-190	420	470	980
Stabilizer Trim Inoperative (Flaps 15)	VREF15+10	120	-100	190	-190	300	360	730
Trailing Edge Flap Asymmetry (1 =< Flap Lever =< 15)	VREF40+30	120	-100	200	-200	290	400	820
Trailing Edge Flap Asymmetry (Flap Lever 15 or 25)	VREF15	100	-90	170	-170	290	290	590
Trailing Edge Flap Asymmetry (Flap Lever 30)	VREF30	90	-80	160	-160	270	220	440
Trailing Edge Flap Disagree (1 =< Indicated Flap < 15)	VREF40+30	120	-100	200	-200	290	400	820
Trailing Edge Flap Disagree (15 =< Indicated Flap < 30)	VREF15	100	-90	170	-170	290	290	590
Trailing Edge Flap Disagree (30 =< Indicated Flap < 40)	VREF30	90	-80	160	-160	270	220	440
Trailing Edge Flaps Up Landing (Flaps Up)	VREF40+40	130	-110	220	-220	300	460	950

Actual (unfactored) distances are shown.

Includes a distance from runway threshold to touchdown associated with a flare time of 7 seconds.

Assumes maximum autobrake setting and maximum available reverse thrust.

For "One Engine Inoperative Landing - Flaps 30" autoland, increase the landing distance by 750 feet.

Continued on next page

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

737-8 Non-Normal Configuration Landing Distances - Dry Runway - 1 Position Tailskid
LEAP-1B Engines

LANDING DISTANCES AND ADJUSTMENTS (METER)								
QRH NON-NORMAL CHECKLIST	REFERENCE SPEED (VREF)	REFERENCE DISTANCE FOR 65000 KGS	WEIGHT ADJUSTMENT PER 5000 KILOGRAMS		ALTITUDE ADJUSTMENT PER 1000 FT ABOVE SEA LEVEL *		RUNWAY WIND ADJUSTMENT PER 10 KNOTS	
			ABOVE 65000 KGS	BELOW 65000 KGS	STD	HIGH	HEAD WIND	TAIL WIND
Airspeed Unreliable (Flaps 15)	VREF15	1575	85	-90	40	50	-65	210
Airspeed Unreliable (Flaps 30)	VREF30	1475	75	-85	40	50	-65	205
Airspeed Unreliable (Flaps 40)	VREF40	1375	90	-80	40	50	-60	200
All Flaps Up Landing (Flaps UP)	VREF40+55	2065	125	-95	55	75	-75	235
ANTISKID INOPERATIVE (Flaps 15)	VREF15	1955	120	-130	60	75	-100	345
ANTISKID INOPERATIVE (Flaps 30)	VREF30	1800	110	-115	55	70	-95	330
ANTISKID INOPERATIVE (Flaps 40)	VREF40	1690	120	-110	55	70	-95	325
Jammed or Restricted Flight Controls (Flaps 15)	VREF15+10	1575	85	-90	40	50	-65	210
LEADING EDGE FLAPS (Flaps 15)	VREF15+15	1650	85	-95	45	55	-65	215
LOSS OF BOTH ENGINE DRIVEN GENERATORS (Flaps 15)	VREF15	2295	150	-160	75	100	-115	385
LOSS OF BOTH ENGINE DRIVEN GENERATORS (Flaps 30)	VREF30	2145	140	-145	70	90	-110	375
LOSS OF BOTH ENGINE DRIVEN GENERATORS (Flaps 40)	VREF40	2020	160	-140	70	90	-110	370
LOSS OF SYSTEM A (Flaps 15)	VREF15	1460	85	-90	40	50	-65	205
LOSS OF SYSTEM A (Flaps 30)	VREF30	1355	80	-85	35	45	-60	200
LOSS OF SYSTEM A (Flaps 40)	VREF40	1260	95	-80	35	45	-60	195
LOSS OF SYSTEM A AND SYSTEM B (Flaps 15)	VREF15	1865	115	-115	55	70	-85	270
LOSS OF SYSTEM B (Flaps 15)	VREF15	1470	90	-90	40	50	-65	220
MANUAL REVERSION (Flaps 15)	VREF15	1865	115	-115	55	70	-85	270

Actual (unfactored) distances are shown.

Includes a distance from runway threshold to touchdown associated with a flare time of 7 seconds.

Assumes maximum autobrake setting and maximum available reverse thrust.

Manual braking is assumed for ANTISKID INOPERATIVE, LOSS OF BOTH ENGINE DRIVEN GENERATORS, LOSS OF SYSTEM B, LOSS OF SYSTEM A AND SYSTEM B and MANUAL REVERSION since autobrakes are inoperative.

*For landing distance at or below 8000 ft pressure altitude, apply the STD altitude adjustment.

For altitudes higher than 8000 ft, first apply the STD altitude adjustment to 8000 ft, then apply the HIGH altitude adjustment for the incremental altitude above 8000 ft.

Continued on next page

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

737-8 Non-Normal Configuration Landing Distances - Dry Runway - 1 Position Tailskid
LEAP-1B Engines

LANDING DISTANCES AND ADJUSTMENTS (METER)								
QRH NON-NORMAL CHECKLIST	REFERENCE SPEED (VREF)	RUNWAY SLOPE ADJUSTMENT PER 1%		TEMPERATURE ADJUSTMENT PER		APPROACH SPEED ADJUSTMENT PER 5 KNOTS ABOVE V_{REF}	REVERSE THRUST ADJUSTMENT	
		DOWN HILL	UP HILL	10° C ABOVE ISA	10° C BELOW ISA		ONE REVERSER	NO REVERSER
Airspeed Unreliable (Flaps 15)	VREF15	0	0	45	-45	N/A	5	10
Airspeed Unreliable (Flaps 30)	VREF30	5	0	40	-40	N/A	0	10
Airspeed Unreliable (Flaps 40)	VREF40	5	0	35	-35	N/A	0	10
All Flaps Up Landing (Flaps UP)	VREF40+55	5	0	55	-60	85	5	15
ANTISKID INOPERATIVE (Flaps 15)	VREF15	60	-50	55	-55	80	155	345
ANTISKID INOPERATIVE (Flaps 30)	VREF30	55	-45	50	-50	75	125	275
ANTISKID INOPERATIVE (Flaps 40)	VREF40	55	-45	45	-50	75	110	240
Jammed or Restricted Flight Controls (Flaps 15)	VREF15+10	0	0	45	-45	75	5	10
LEADING EDGE FLAPS (Flaps 15)	VREF15+15	5	0	45	-45	85	5	10
LOSS OF BOTH ENGINE DRIVEN GENERATORS (Flaps 15)	VREF15	90	-75	65	-70	115	280	655
LOSS OF BOTH ENGINE DRIVEN GENERATORS (Flaps 30)	VREF30	85	-70	60	-65	110	235	545
LOSS OF BOTH ENGINE DRIVEN GENERATORS (Flaps 40)	VREF40	85	-70	60	-60	110	210	475
LOSS OF SYSTEM A (Flaps 15)	VREF15	10	-5	40	-40	80	25	45
LOSS OF SYSTEM A (Flaps 30)	VREF30	10	-5	35	-35	75	20	35
LOSS OF SYSTEM A (Flaps 40)	VREF40	10	-5	35	-35	75	20	30
LOSS OF SYSTEM A AND SYSTEM B (Flaps 15)	VREF15	50	-45	50	-50	115	20	95
LOSS OF SYSTEM B (Flaps 15)	VREF15	25	-20	40	-40	70	45	75
MANUAL REVERSION (Flaps 15)	VREF15	50	-45	50	-50	115	20	95

Actual (unfactored) distances are shown.

Includes a distance from runway threshold to touchdown associated with a flare time of 7 seconds.

Assumes maximum autobrake setting and maximum available reverse thrust.

Manual braking is assumed for ANTISKID INOPERATIVE, LOSS OF BOTH ENGINE DRIVEN GENERATORS, LOSS OF SYSTEM B, LOSS OF SYSTEM A AND SYSTEM B and MANUAL REVERSION since autobrakes are inoperative.

Continued on next page

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

737-8 Non-Normal Configuration Landing Distances - Dry Runway - 1 Position Tailskid
LEAP-1B Engines

LANDING DISTANCES AND ADJUSTMENTS (METER)								
QRH NON-NORMAL CHECKLIST	REFERENCE SPEED (VREF)	REFERENCE DISTANCE FOR 65000 KGS	WEIGHT ADJUSTMENT PER 5000 KILOGRAMS		ALTITUDE ADJUSTMENT PER 1000 FT ABOVE SEA LEVEL *		RUNWAY WIND ADJUSTMENT PER 10 KNOTS	
			ABOVE 65000 KGS	BELOW 65000 KGS	STD	HIGH	HEAD WIND	TAIL WIND
One Engine Inoperative Landing (Flaps 15)	VREF15	1430	80	-90	35	45	-65	205
One Engine Inoperative Landing (Flaps 30)	VREF30	1335	75	-80	35	45	-60	200
SPOILERS (Flaps 15)	VREF15	1480	90	-95	40	55	-65	205
SPOILERS (Flaps 30)	VREF30	1395	85	-90	40	50	-65	205
SPOILERS (Flaps 40)	VREF40	1315	100	-85	40	50	-60	200
Stabilizer Trim Inoperative (Flaps 15)	VREF15+10	1575	85	-90	40	50	-65	210
Trailing Edge Flap Asymmetry (1 =< Flap Lever =< 15)	VREF40+30	1665	95	-85	45	60	-65	215
Trailing Edge Flap Asymmetry (Flap Lever 15 or 25)	VREF15	1430	80	-90	35	45	-65	205
Trailing Edge Flap Asymmetry (Flap Lever 30)	VREF30	1335	70	-80	35	45	-60	195
Trailing Edge Flap Disagree (1 =< Indicated Flap < 15)	VREF40+30	1665	95	-85	45	60	-65	215
Trailing Edge Flap Disagree (15 =< Indicated Flap < 30)	VREF15	1430	80	-90	35	45	-65	205
Trailing Edge Flap Disagree (30 =< Indicated Flap < 40)	VREF30	1335	70	-80	35	45	-60	195
Trailing Edge Flaps Up Landing (Flaps Up)	VREF40+40	1820	105	-90	50	65	-70	225

Actual (unfactored) distances are shown.

Includes a distance from runway threshold to touchdown associated with a flare time of 7 seconds.

Assumes maximum autobrake setting and maximum available reverse thrust.

For "One Engine Inoperative Landing - Flaps 30" autoland, increase the landing distance by 230 meters.

*For landing distance at or below 8000 ft pressure altitude, apply the STD altitude adjustment.

For altitudes higher than 8000 ft, first apply the STD altitude adjustment to 8000 ft,

then apply the HIGH altitude adjustment for the incremental altitude above 8000 ft.

Continued on next page

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

737-8 Non-Normal Configuration Landing Distances - Dry Runway - 1 Position Tailskid
LEAP-1B Engines

LANDING DISTANCES AND ADJUSTMENTS (METER)								
QRH NON-NORMAL CHECKLIST	REFERENCE SPEED (VREF)	RUNWAY SLOPE ADJUSTMENT PER 1%		TEMPERATURE ADJUSTMENT PER		APPROACH SPEED ADJUSTMENT PER 5 KNOTS ABOVE V_{REF}	REVERSE THRUST ADJUSTMENT	
		DOWN HILL	UP HILL	10° C ABOVE ISA	10° C BELOW ISA		ONE REVERSER	NO REVERSER
One Engine Inoperative Landing (Flaps 15)	VREF15	5	0	40	-40	75	N/A	5
One Engine Inoperative Landing (Flaps 30)	VREF30	5	0	35	-35	70	N/A	5
SPOILERS (Flaps 15)	VREF15	15	-10	40	-40	85	55	85
SPOILERS (Flaps 30)	VREF30	15	-10	40	-40	85	50	80
SPOILERS (Flaps 40)	VREF40	15	-15	35	-35	90	45	70
Stabilizer Trim Inoperative (Flaps 15)	VREF15+10	0	0	45	-45	75	5	10
Trailing Edge Flap Asymmetry (1 =< Flap Lever =< 15)	VREF40+30	0	0	45	-45	80	5	10
Trailing Edge Flap Asymmetry (Flap Lever 15 or 25)	VREF15	0	0	40	-40	75	5	10
Trailing Edge Flap Asymmetry (Flap Lever 30)	VREF30	0	0	35	-35	70	0	10
Trailing Edge Flap Disagree (1 =< Indicated Flap < 15)	VREF40+30	0	0	45	-45	80	5	10
Trailing Edge Flap Disagree (15 =< Indicated Flap < 30)	VREF15	0	0	40	-40	75	5	10
Trailing Edge Flap Disagree (30 =< Indicated Flap < 40)	VREF30	0	0	35	-35	70	0	10
Trailing Edge Flaps Up Landing (Flaps Up)	VREF40+40	0	0	50	-50	80	5	10

Actual (unfactored) distances are shown.

Includes a distance from runway threshold to touchdown associated with a flare time of 7 seconds.

Assumes maximum autobrake setting and maximum available reverse thrust.

For "One Engine Inoperative Landing - Flaps 30" autoland, increase the landing distance by 230 meters.

Continued on next page

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

737-8 Non-Normal Configuration Landing Distances - Wet Runway - 1 Position Tailskid
LEAP-1B Engines

LANDING DISTANCES AND ADJUSTMENTS (METER)								
QRH NON-NORMAL CHECKLIST	REFERENCE SPEED (VREF)	REFERENCE DISTANCE FOR 65000 KGS	WEIGHT ADJUSTMENT PER 5000 KILOGRAMS		ALTITUDE ADJUSTMENT PER 1000 FT ABOVE SEA LEVEL *		RUNWAY WIND ADJUSTMENT PER 10 KNOTS	
			ABOVE 65000 KGS	BELOW 65000 KGS	STD	HIGH	HEAD WIND	TAIL WIND
Airspeed Unreliable (Flaps 15)	VREF15	1790	125	-120	65	85	-90	315
Airspeed Unreliable (Flaps 30)	VREF30	1645	110	-110	55	75	-85	300
Airspeed Unreliable (Flaps 40)	VREF40	1545	125	-100	55	75	-80	290
All Flaps Up Landing (Flaps UP)	VREF40+55	2350	160	-140	90	120	-105	365
ANTISKID INOPERATIVE (Flaps 15)	VREF15	2275	175	-170	85	120	-135	485
ANTISKID INOPERATIVE (Flaps 30)	VREF30	2065	155	-150	75	105	-125	460
ANTISKID INOPERATIVE (Flaps 40)	VREF40	1920	165	-140	75	100	-120	445
Jammed or Restricted Flight Controls (Flaps 15)	VREF15+10	1790	125	-120	65	85	-90	315
LEADING EDGE FLAPS (Flaps 15)	VREF15+15	1915	130	-130	70	95	-95	330
LOSS OF BOTH ENGINE DRIVEN GENERATORS (Flaps 15)	VREF15	2690	225	-210	110	155	-150	545
LOSS OF BOTH ENGINE DRIVEN GENERATORS (Flaps 30)	VREF30	2470	200	-190	100	135	-145	525
LOSS OF BOTH ENGINE DRIVEN GENERATORS (Flaps 40)	VREF40	2300	215	-175	95	130	-140	510
LOSS OF SYSTEM A (Flaps 15)	VREF15	1845	145	-140	70	100	-100	355
LOSS OF SYSTEM A (Flaps 30)	VREF30	1655	125	-120	60	85	-95	330
LOSS OF SYSTEM A (Flaps 40)	VREF40	1525	140	-110	60	80	-90	315
LOSS OF SYSTEM A AND SYSTEM B (Flaps 15)	VREF15	2510	190	-185	95	135	-135	465
LOSS OF SYSTEM B (Flaps 15)	VREF15	1860	140	-135	70	95	-105	365
MANUAL REVERSION (Flaps 15)	VREF15	2510	190	-185	95	135	-135	465

Actual (unfactored) distances are shown.

Includes a distance from runway threshold to touchdown associated with a flare time of 7 seconds.

Assumes maximum autobrake setting and maximum available reverse thrust.

Manual braking is assumed for ANTISKID INOPERATIVE, LOSS OF BOTH ENGINE DRIVEN GENERATORS, LOSS OF SYSTEM B, LOSS OF SYSTEM A AND SYSTEM B and MANUAL REVERSION since autobrakes are inoperative.

*For landing distance at or below 8000 ft pressure altitude, apply the STD altitude adjustment.

For altitudes higher than 8000 ft, first apply the STD altitude adjustment to 8000 ft, then apply the HIGH altitude adjustment for the incremental altitude above 8000 ft.

Continued on next page

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

737-8 Non-Normal Configuration Landing Distances - Wet Runway - 1 Position Tailskid
LEAP-1B Engines

LANDING DISTANCES AND ADJUSTMENTS (METER)								
QRH NON-NORMAL CHECKLIST	REFERENCE SPEED (VREF)	RUNWAY SLOPE ADJUSTMENT PER 1%		TEMPERATURE ADJUSTMENT PER		APPROACH SPEED ADJUSTMENT PER 5 KNOTS ABOVE V_{REF}	REVERSE THRUST ADJUSTMENT	
		DOWN HILL	UP HILL	10° C ABOVE ISA	10° C BELOW ISA		ONE REVERSER	NO REVERSER
Airspeed Unreliable (Flaps 15)	VREF15	35	-30	60	-60	N/A	110	220
Airspeed Unreliable (Flaps 30)	VREF30	30	-25	55	-55	N/A	85	165
Airspeed Unreliable (Flaps 40)	VREF40	30	-25	50	-50	N/A	70	135
All Flaps Up Landing (Flaps UP)	VREF40+55	50	-40	80	-80	95	195	405
ANTISKID INOPERATIVE (Flaps 15)	VREF15	95	-75	75	-75	100	300	765
ANTISKID INOPERATIVE (Flaps 30)	VREF30	85	-65	70	-70	95	225	555
ANTISKID INOPERATIVE (Flaps 40)	VREF40	80	-65	60	-60	95	190	460
Jammed or Restricted Flight Controls (Flaps 15)	VREF15+10	35	-30	60	-60	90	110	220
LEADING EDGE FLAPS (Flaps 15)	VREF15+15	40	-35	65	-65	95	130	270
LOSS OF BOTH ENGINE DRIVEN GENERATORS (Flaps 15)	VREF15	145	-115	90	-90	135	545	1585
LOSS OF BOTH ENGINE DRIVEN GENERATORS (Flaps 30)	VREF30	135	-105	85	-85	130	435	1180
LOSS OF BOTH ENGINE DRIVEN GENERATORS (Flaps 40)	VREF40	125	-100	75	-75	125	365	965
LOSS OF SYSTEM A (Flaps 15)	VREF15	60	-50	65	-65	115	155	290
LOSS OF SYSTEM A (Flaps 30)	VREF30	50	-45	55	-55	105	110	195
LOSS OF SYSTEM A (Flaps 40)	VREF40	45	-40	50	-50	105	85	150
LOSS OF SYSTEM A AND SYSTEM B (Flaps 15)	VREF15	120	-95	85	-85	155	140	590
LOSS OF SYSTEM B (Flaps 15)	VREF15	50	-45	65	-65	100	155	300
MANUAL REVERSION (Flaps 15)	VREF15	120	-95	85	-85	155	140	590

Actual (unfactored) distances are shown.

Includes a distance from runway threshold to touchdown associated with a flare time of 7 seconds.

Assumes maximum autobrake setting and maximum available reverse thrust.

Manual braking is assumed for ANTISKID INOPERATIVE, LOSS OF BOTH ENGINE DRIVEN GENERATORS, LOSS OF SYSTEM B, LOSS OF SYSTEM A AND SYSTEM B and MANUAL REVERSION since autobrakes are inoperative.

Continued on next page

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

737-8 Non-Normal Configuration Landing Distances - Wet Runway - 1 Position Tailskid
LEAP-1B Engines

LANDING DISTANCES AND ADJUSTMENTS (METER)								
QRH NON-NORMAL CHECKLIST	REFERENCE SPEED (VREF)	REFERENCE DISTANCE FOR 65000 KGS	WEIGHT ADJUSTMENT PER 5000 KILOGRAMS		ALTITUDE ADJUSTMENT PER 1000 FT ABOVE SEA LEVEL *		RUNWAY WIND ADJUSTMENT PER 10 KNOTS	
			ABOVE 65000 KGS	BELOW 65000 KGS	STD	HIGH	HEAD WIND	TAIL WIND
One Engine Inoperative Landing (Flaps 15)	VREF15	1670	125	-120	60	80	-90	325
One Engine Inoperative Landing (Flaps 30)	VREF30	1525	110	-110	55	70	-85	305
SPOILERS (Flaps 15)	VREF15	1920	155	-150	80	110	-105	365
SPOILERS (Flaps 30)	VREF30	1780	140	-135	70	100	-100	350
SPOILERS (Flaps 40)	VREF40	1660	165	-125	70	95	-95	340
Stabilizer Trim Inoperative (Flaps 15)	VREF15+10	1790	125	-120	65	85	-90	315
Trailing Edge Flap Asymmetry (1 =< Flap Lever =< 15)	VREF40+30	1865	135	-115	70	95	-90	325
Trailing Edge Flap Asymmetry (Flap Lever 15 or 25)	VREF15	1615	115	-115	55	80	-85	305
Trailing Edge Flap Asymmetry (Flap Lever 30)	VREF30	1485	105	-105	50	70	-80	285
Trailing Edge Flap Disagree (1 =< Indicated Flap < 15)	VREF40+30	1865	135	-115	70	95	-90	325
Trailing Edge Flap Disagree (15 =< Indicated Flap < 30)	VREF15	1615	115	-115	55	80	-85	305
Trailing Edge Flap Disagree (30 =< Indicated Flap < 40)	VREF30	1485	105	-105	50	70	-80	285
Trailing Edge Flaps Up Landing (Flaps Up)	VREF40+40	2030	145	-125	75	105	-95	335

Actual (unfactored) distances are shown.

Includes a distance from runway threshold to touchdown associated with a flare time of 7 seconds.

Assumes maximum autobrake setting and maximum available reverse thrust.

For "One Engine Inoperative Landing - Flaps 30" autoland, increase the landing distance by 230 meters.

*For landing distance at or below 8000 ft pressure altitude, apply the STD altitude adjustment.

For altitudes higher than 8000 ft, first apply the STD altitude adjustment to 8000 ft,

then apply the HIGH altitude adjustment for the incremental altitude above 8000 ft.

Continued on next page

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

Section Non-Normal Configuration Landing Distances, continued from previous page A170245

737-8 Non-Normal Configuration Landing Distances - Wet Runway - 1 Position Tailskid
LEAP-1B Engines

LANDING DISTANCES AND ADJUSTMENTS (METER)								
QRH NON-NORMAL CHECKLIST	REFERENCE SPEED (VREF)	RUNWAY SLOPE ADJUSTMENT PER 1%		TEMPERATURE ADJUSTMENT PER		APPROACH SPEED ADJUSTMENT PER 5 KNOTS ABOVE V_{REF}	REVERSE THRUST ADJUSTMENT	
		DOWN HILL	UP HILL	10° C ABOVE ISA	10° C BELOW ISA		ONE REVERSER	NO REVERSER
One Engine Inoperative Landing (Flaps 15)	VREF15	35	-30	60	-55	95	N/A	100
One Engine Inoperative Landing (Flaps 30)	VREF30	30	-25	50	-50	90	N/A	70
SPOILERS (Flaps 15)	VREF15	75	-60	70	-70	130	210	460
SPOILERS (Flaps 30)	VREF30	70	-55	65	-65	130	170	365
SPOILERS (Flaps 40)	VREF40	65	-55	60	-60	130	145	300
Stabilizer Trim Inoperative (Flaps 15)	VREF15+10	35	-30	60	-60	90	110	220
Trailing Edge Flap Asymmetry (1 =< Flap Lever =< 15)	VREF40+30	35	-30	60	-60	90	120	250
Trailing Edge Flap Asymmetry (Flap Lever 15 or 25)	VREF15	30	-25	55	-55	90	90	180
Trailing Edge Flap Asymmetry (Flap Lever 30)	VREF30	25	-25	50	-50	85	70	135
Trailing Edge Flap Disagree (1 =< Indicated Flap < 15)	VREF40+30	35	-30	60	-60	90	120	250
Trailing Edge Flap Disagree (15 =< Indicated Flap < 30)	VREF15	30	-25	55	-55	90	90	180
Trailing Edge Flap Disagree (30 =< Indicated Flap < 40)	VREF30	25	-25	50	-50	85	70	135
Trailing Edge Flaps Up Landing (Flaps Up)	VREF40+40	40	-35	70	-70	90	140	290

Actual (unfactored) distances are shown.

Includes a distance from runway threshold to touchdown associated with a flare time of 7 seconds.

Assumes maximum autobrake setting and maximum available reverse thrust.

For "One Engine Inoperative Landing - Flaps 30" autoland, increase the landing distance by 230 meters.

Continued on next page

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

737-8 Non-Normal Configuration Landing Distances - Dry Runway - 2 Position Tailskid
LEAP-1B Engines

LANDING DISTANCES AND ADJUSTMENTS (FEET)								
QRH NON-NORMAL CHECKLIST	REFERENCE SPEED (VREF)	REFERENCE DISTANCE FOR 150000 LBS	WEIGHT ADJUSTMENT PER 10000 POUNDS		ALTITUDE ADJUSTMENT PER 1000 FT ABOVE SEA LEVEL *		RUNWAY WIND ADJUSTMENT PER 10 KNOTS	
			ABOVE 150000 LBS	BELOW 150000 LBS	STD	HIGH	HEAD WIND	TAIL WIND
Airspeed Unreliable (Flaps 15)	VREF15	5260	240	-260	130	180	-210	690
Airspeed Unreliable (Flaps 30)	VREF30	4830	220	-240	120	160	-200	660
Airspeed Unreliable (Flaps 40)	VREF40	4650	270	-230	130	160	-200	650
All Flaps Up Landing (Flaps UP)	VREF40+55	6930	380	-280	190	240	-240	780
ANTISKID INOPERATIVE (Flaps 15)	VREF15	6560	350	-370	190	260	-330	1120
ANTISKID INOPERATIVE (Flaps 30)	VREF30	5950	320	-330	170	230	-310	1080
ANTISKID INOPERATIVE (Flaps 40)	VREF40	5730	370	-320	180	220	-310	1070
Jammed or Restricted Flight Controls (Flaps 15)	VREF15+10	5260	240	-260	130	180	-210	690
LEADING EDGE FLAPS (Flaps 15)	VREF15+15	5500	250	-270	140	190	-220	700
LOSS OF BOTH ENGINE DRIVEN GENERATORS (Flaps 15)	VREF15	7710	450	-450	250	340	-370	1250
LOSS OF BOTH ENGINE DRIVEN GENERATORS (Flaps 30)	VREF30	7070	400	-410	220	300	-360	1210
LOSS OF BOTH ENGINE DRIVEN GENERATORS (Flaps 40)	VREF40	6860	480	-400	230	290	-360	1210
LOSS OF SYSTEM A (Flaps 15)	VREF15	4870	260	-260	130	170	-210	670
LOSS OF SYSTEM A (Flaps 30)	VREF30	4430	230	-230	110	150	-200	640
LOSS OF SYSTEM A (Flaps 40)	VREF40	4270	280	-230	120	150	-200	630
LOSS OF SYSTEM A AND SYSTEM B (Flaps 15)	VREF15	6220	340	-330	170	240	-270	880
LOSS OF SYSTEM B (Flaps 15)	VREF15	4930	270	-260	130	180	-220	720
MANUAL REVERSION (Flaps 15)	VREF15	6220	340	-330	170	240	-270	880

Actual (unfactored) distances are shown.

Includes a distance from runway threshold to touchdown associated with a flare time of 7 seconds.

Assumes maximum autobrake setting and maximum available reverse thrust.

Manual braking is assumed for ANTISKID INOPERATIVE, LOSS OF BOTH ENGINE DRIVEN GENERATORS, LOSS OF SYSTEM B, LOSS OF SYSTEM A AND SYSTEM B and MANUAL REVERSION since autobrakes are inoperative.

*For landing distance at or below 8000 ft pressure altitude, apply the STD altitude adjustment.

For altitudes higher than 8000 ft, first apply the STD altitude adjustment to 8000 ft, then apply the HIGH altitude adjustment for the incremental altitude above 8000 ft.

Continued on next page

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

737-8 Non-Normal Configuration Landing Distances - Dry Runway - 2 Position Tailskid
LEAP-1B Engines

LANDING DISTANCES AND ADJUSTMENTS (FEET)								
QRH NON-NORMAL CHECKLIST	REFERENCE SPEED (VREF)	RUNWAY SLOPE ADJUSTMENT PER 1%		TEMPERATURE ADJUSTMENT PER		APPROACH SPEED ADJUSTMENT PER 5 KNOTS ABOVE V_{REF}	REVERSE THRUST ADJUSTMENT	
		DOWN HILL	UP HILL	10° C ABOVE ISA	10° C BELOW ISA		ONE REVERSER	NO REVERSER
Airspeed Unreliable (Flaps 15)	VREF15	10	0	140	-140	N/A	10	30
Airspeed Unreliable (Flaps 30)	VREF30	10	0	130	-130	N/A	10	30
Airspeed Unreliable (Flaps 40)	VREF40	20	-10	120	-120	N/A	10	30
All Flaps Up Landing (Flaps UP)	VREF40+55	10	0	190	-190	280	10	40
ANTISKID INOPERATIVE (Flaps 15)	VREF15	190	-160	180	-180	270	500	1120
ANTISKID INOPERATIVE (Flaps 30)	VREF30	180	-140	160	-160	250	390	870
ANTISKID INOPERATIVE (Flaps 40)	VREF40	170	-140	150	-160	250	360	780
Jammed or Restricted Flight Controls (Flaps 15)	VREF15+10	10	0	140	-140	250	10	30
LEADING EDGE FLAPS (Flaps 15)	VREF15+15	20	0	140	-150	270	10	40
LOSS OF BOTH ENGINE DRIVEN GENERATORS (Flaps 15)	VREF15	290	-240	220	-220	370	900	2090
LOSS OF BOTH ENGINE DRIVEN GENERATORS (Flaps 30)	VREF30	270	-220	200	-200	360	740	1690
LOSS OF BOTH ENGINE DRIVEN GENERATORS (Flaps 40)	VREF40	280	-220	190	-190	370	680	1550
LOSS OF SYSTEM A (Flaps 15)	VREF15	40	-20	130	-130	250	80	140
LOSS OF SYSTEM A (Flaps 30)	VREF30	30	-20	110	-120	240	60	100
LOSS OF SYSTEM A (Flaps 40)	VREF40	40	-20	110	-110	240	60	100
LOSS OF SYSTEM A AND SYSTEM B (Flaps 15)	VREF15	160	-140	170	-160	370	50	300
LOSS OF SYSTEM B (Flaps 15)	VREF15	80	-70	130	-130	220	140	240
MANUAL REVERSION (Flaps 15)	VREF15	160	-140	170	-160	370	50	300

Actual (unfactored) distances are shown.

Includes a distance from runway threshold to touchdown associated with a flare time of 7 seconds.

Assumes maximum autobrake setting and maximum available reverse thrust.

Manual braking is assumed for ANTISKID INOPERATIVE, LOSS OF BOTH ENGINE DRIVEN GENERATORS, LOSS OF SYSTEM B, LOSS OF SYSTEM A AND SYSTEM B and MANUAL REVERSION since autobrakes are inoperative.

Continued on next page

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

737-8 Non-Normal Configuration Landing Distances - Dry Runway - 2 Position Tailskid
LEAP-1B Engines

LANDING DISTANCES AND ADJUSTMENTS (FEET)								
QRH NON-NORMAL CHECKLIST	REFERENCE SPEED (VREF)	REFERENCE DISTANCE FOR 150000 LBS	WEIGHT ADJUSTMENT PER 10000 POUNDS		ALTITUDE ADJUSTMENT PER 1000 FT ABOVE SEA LEVEL *		RUNWAY WIND ADJUSTMENT PER 10 KNOTS	
			ABOVE 150000 LBS	BELOW 150000 LBS	STD	HIGH	HEAD WIND	TAIL WIND
One Engine Inoperative Landing (Flaps 15)	VREF15	4780	240	-250	120	160	-200	660
One Engine Inoperative Landing (Flaps 30)	VREF30	4370	210	-230	110	140	-200	640
SPOILERS (Flaps 15)	VREF15	4940	270	-270	130	180	-210	670
SPOILERS (Flaps 30)	VREF30	4560	240	-250	130	170	-200	660
SPOILERS (Flaps 40)	VREF40	4450	300	-250	130	170	-200	660
Stabilizer Trim Inoperative (Flaps 15)	VREF15+10	5260	240	-260	130	180	-210	690
Trailing Edge Flap Asymmetry (1 =< Flap Lever =< 15)	VREF40+30	5610	290	-260	150	190	-220	710
Trailing Edge Flap Asymmetry (Flap Lever 15 or 25)	VREF15	4780	230	-250	120	160	-200	660
Trailing Edge Flap Asymmetry (Flap Lever 30)	VREF30	4370	210	-230	110	140	-200	640
Trailing Edge Flap Disagree (1 =< Indicated Flap < 15)	VREF40+30	5610	290	-260	150	190	-220	710
Trailing Edge Flap Disagree (15 =< Indicated Flap < 30)	VREF15	4780	230	-250	120	160	-200	660
Trailing Edge Flap Disagree (30 =< Indicated Flap < 40)	VREF30	4370	210	-230	110	140	-200	640
Trailing Edge Flaps Up Landing (Flaps Up)	VREF40+40	6120	320	-270	170	210	-230	740

Actual (unfactored) distances are shown.

Includes a distance from runway threshold to touchdown associated with a flare time of 7 seconds.

Assumes maximum autobrake setting and maximum available reverse thrust.

For "One Engine Inoperative Landing - Flaps 30" autoland, increase the landing distance by 700 feet.

*For landing distance at or below 8000 ft pressure altitude, apply the STD altitude adjustment.

For altitudes higher than 8000 ft, first apply the STD altitude adjustment to 8000 ft,

then apply the HIGH altitude adjustment for the incremental altitude above 8000 ft.

Continued on next page

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

737-8 Non-Normal Configuration Landing Distances - Dry Runway - 2 Position Tailskid
LEAP-1B Engines

LANDING DISTANCES AND ADJUSTMENTS (FEET)								
QRH NON-NORMAL CHECKLIST	REFERENCE SPEED (VREF)	RUNWAY SLOPE ADJUSTMENT PER 1%		TEMPERATURE ADJUSTMENT PER		APPROACH SPEED ADJUSTMENT PER 5 KNOTS ABOVE V_{REF}	REVERSE THRUST ADJUSTMENT	
		DOWN HILL	UP HILL	10° C ABOVE ISA	10° C BELOW ISA		ONE REVERSER	NO REVERSER
One Engine Inoperative Landing (Flaps 15)	VREF15	10	-10	120	-130	240	N/A	20
One Engine Inoperative Landing (Flaps 30)	VREF30	10	0	110	-110	230	N/A	10
SPOILERS (Flaps 15)	VREF15	50	-30	130	-130	270	170	270
SPOILERS (Flaps 30)	VREF30	50	-40	120	-120	270	150	240
SPOILERS (Flaps 40)	VREF40	50	-40	120	-120	290	140	220
Stabilizer Trim Inoperative (Flaps 15)	VREF15+10	10	0	140	-140	250	10	30
Trailing Edge Flap Asymmetry (1 =< Flap Lever =< 15)	VREF40+30	10	0	150	-150	260	10	30
Trailing Edge Flap Asymmetry (Flap Lever 15 or 25)	VREF15	10	0	120	-130	240	10	30
Trailing Edge Flap Asymmetry (Flap Lever 30)	VREF30	10	0	110	-110	230	10	20
Trailing Edge Flap Disagree (1 =< Indicated Flap < 15)	VREF40+30	10	0	150	-150	260	10	30
Trailing Edge Flap Disagree (15 =< Indicated Flap < 30)	VREF15	10	0	120	-130	240	10	30
Trailing Edge Flap Disagree (30 =< Indicated Flap < 40)	VREF30	10	0	110	-110	230	10	20
Trailing Edge Flaps Up Landing (Flaps Up)	VREF40+40	10	0	160	-170	270	10	30

Actual (unfactored) distances are shown.

Includes a distance from runway threshold to touchdown associated with a flare time of 7 seconds.

Assumes maximum autobrake setting and maximum available reverse thrust.

For "One Engine Inoperative Landing - Flaps 30" autoland, increase the landing distance by 700 feet.

Continued on next page

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

737-8 Non-Normal Configuration Landing Distances - Wet Runway - 2 Position Tailskid
LEAP-1B Engines

LANDING DISTANCES AND ADJUSTMENTS (FEET)								
QRH NON-NORMAL CHECKLIST	REFERENCE SPEED (VREF)	REFERENCE DISTANCE FOR 150000 LBS	WEIGHT ADJUSTMENT PER 10000 POUNDS		ALTITUDE ADJUSTMENT PER 1000 FT ABOVE SEA LEVEL *		RUNWAY WIND ADJUSTMENT PER 10 KNOTS	
			ABOVE 150000 LBS	BELOW 150000 LBS	STD	HIGH	HEAD WIND	TAIL WIND
Airspeed Unreliable (Flaps 15)	VREF15	6010	360	-350	210	300	-290	1030
Airspeed Unreliable (Flaps 30)	VREF30	5430	320	-310	180	250	-270	970
Airspeed Unreliable (Flaps 40)	VREF40	5250	370	-300	180	250	-270	950
All Flaps Up Landing (Flaps UP)	VREF40+55	7960	480	-410	300	400	-350	1200
ANTISKID INOPERATIVE (Flaps 15)	VREF15	7690	520	-480	280	400	-430	1580
ANTISKID INOPERATIVE (Flaps 30)	VREF30	6860	450	-420	240	340	-400	1490
ANTISKID INOPERATIVE (Flaps 40)	VREF40	6550	500	-400	240	320	-390	1460
Jammed or Restricted Flight Controls (Flaps 15)	VREF15+10	6010	360	-350	210	300	-290	1030
LEADING EDGE FLAPS (Flaps 15)	VREF15+15	6430	380	-370	220	320	-310	1070
LOSS OF BOTH ENGINE DRIVEN GENERATORS (Flaps 15)	VREF15	9110	660	-600	360	520	-490	1780
LOSS OF BOTH ENGINE DRIVEN GENERATORS (Flaps 30)	VREF30	8210	580	-530	320	450	-470	1690
LOSS OF BOTH ENGINE DRIVEN GENERATORS (Flaps 40)	VREF40	7870	650	-510	320	430	-460	1660
LOSS OF SYSTEM A (Flaps 15)	VREF15	6200	420	-400	230	340	-330	1160
LOSS OF SYSTEM A (Flaps 30)	VREF30	5440	360	-340	200	280	-300	1070
LOSS OF SYSTEM A (Flaps 40)	VREF40	5200	430	-320	200	270	-290	1040
LOSS OF SYSTEM A AND SYSTEM B (Flaps 15)	VREF15	8430	560	-530	310	450	-440	1520
LOSS OF SYSTEM B (Flaps 15)	VREF15	6270	410	-390	220	320	-330	1190
MANUAL REVERSION (Flaps 15)	VREF15	8430	560	-530	310	450	-440	1520

Actual (unfactored) distances are shown.

Includes a distance from runway threshold to touchdown associated with a flare time of 7 seconds.

Assumes maximum autobrake setting and maximum available reverse thrust.

Manual braking is assumed for ANTISKID INOPERATIVE, LOSS OF BOTH ENGINE DRIVEN GENERATORS, LOSS OF SYSTEM B, LOSS OF SYSTEM A AND SYSTEM B and MANUAL REVERSION since autobrakes are inoperative.

*For landing distance at or below 8000 ft pressure altitude, apply the STD altitude adjustment.

For altitudes higher than 8000 ft, first apply the STD altitude adjustment to 8000 ft, then apply the HIGH altitude adjustment for the incremental altitude above 8000 ft.

Continued on next page

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

737-8 Non-Normal Configuration Landing Distances - Wet Runway - 2 Position Tailskid
LEAP-1B Engines

LANDING DISTANCES AND ADJUSTMENTS (FEET)								
QRH NON-NORMAL CHECKLIST	REFERENCE SPEED (VREF)	RUNWAY SLOPE ADJUSTMENT PER 1%		TEMPERATURE ADJUSTMENT PER		APPROACH SPEED ADJUSTMENT PER 5 KNOTS ABOVE V_{REF}	REVERSE THRUST ADJUSTMENT	
		DOWN HILL	UP HILL	10° C ABOVE ISA	10° C BELOW ISA		ONE REVERSER	NO REVERSER
Airspeed Unreliable (Flaps 15)	VREF15	110	-100	190	-190	N/A	350	700
Airspeed Unreliable (Flaps 30)	VREF30	100	-80	170	-170	N/A	260	500
Airspeed Unreliable (Flaps 40)	VREF40	90	-80	160	-160	N/A	230	440
All Flaps Up Landing (Flaps UP)	VREF40+55	160	-140	260	-260	320	630	1320
ANTISKID INOPERATIVE (Flaps 15)	VREF15	310	-250	240	-240	340	960	2450
ANTISKID INOPERATIVE (Flaps 30)	VREF30	270	-210	220	-210	310	710	1750
ANTISKID INOPERATIVE (Flaps 40)	VREF40	260	-210	200	-200	310	630	1500
Jammed or Restricted Flight Controls (Flaps 15)	VREF15+10	110	-100	190	-190	290	350	700
LEADING EDGE FLAPS (Flaps 15)	VREF15+15	130	-110	210	-200	310	420	850
LOSS OF BOTH ENGINE DRIVEN GENERATORS (Flaps 15)	VREF15	470	-370	300	-300	450	1750	5050
LOSS OF BOTH ENGINE DRIVEN GENERATORS (Flaps 30)	VREF30	420	-330	270	-260	420	1350	3660
LOSS OF BOTH ENGINE DRIVEN GENERATORS (Flaps 40)	VREF40	410	-320	250	-250	420	1190	3140
LOSS OF SYSTEM A (Flaps 15)	VREF15	200	-170	210	-210	380	490	910
LOSS OF SYSTEM A (Flaps 30)	VREF30	160	-130	180	-180	350	320	570
LOSS OF SYSTEM A (Flaps 40)	VREF40	150	-130	170	-170	340	280	480
LOSS OF SYSTEM A AND SYSTEM B (Flaps 15)	VREF15	380	-310	280	-270	500	430	1840
LOSS OF SYSTEM B (Flaps 15)	VREF15	170	-140	200	-200	320	490	940
MANUAL REVERSION (Flaps 15)	VREF15	380	-310	280	-270	500	430	1840

Actual (unfactored) distances are shown.

Includes a distance from runway threshold to touchdown associated with a flare time of 7 seconds.

Assumes maximum autobrake setting and maximum available reverse thrust.

Manual braking is assumed for ANTISKID INOPERATIVE, LOSS OF BOTH ENGINE DRIVEN GENERATORS, LOSS OF SYSTEM B, LOSS OF SYSTEM A AND SYSTEM B and MANUAL REVERSION since autobrakes are inoperative.

Continued on next page

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

737-8 Non-Normal Configuration Landing Distances - Wet Runway - 2 Position Tailskid
LEAP-1B Engines

LANDING DISTANCES AND ADJUSTMENTS (FEET)								
QRH NON-NORMAL CHECKLIST	REFERENCE SPEED (VREF)	REFERENCE DISTANCE FOR 150000 LBS	WEIGHT ADJUSTMENT PER 10000 POUNDS		ALTITUDE ADJUSTMENT PER 1000 FT ABOVE SEA LEVEL *		RUNWAY WIND ADJUSTMENT PER 10 KNOTS	
			ABOVE 150000 LBS	BELOW 150000 LBS	STD	HIGH	HEAD WIND	TAIL WIND
One Engine Inoperative Landing (Flaps 15)	VREF15	5620	370	-350	190	280	-300	1060
One Engine Inoperative Landing (Flaps 30)	VREF30	5020	320	-300	170	240	-270	980
SPOILERS (Flaps 15)	VREF15	6450	460	-420	250	370	-340	1190
SPOILERS (Flaps 30)	VREF30	5830	410	-370	220	320	-320	1130
SPOILERS (Flaps 40)	VREF40	5670	500	-370	230	320	-320	1110
Stabilizer Trim Inoperative (Flaps 15)	VREF15+10	6010	360	-350	210	300	-290	1030
Trailing Edge Flap Asymmetry (1 =< Flap Lever =< 15)	VREF40+30	6330	410	-340	230	310	-300	1060
Trailing Edge Flap Asymmetry (Flap Lever 15 or 25)	VREF15	5430	340	-330	180	270	-280	990
Trailing Edge Flap Asymmetry (Flap Lever 30)	VREF30	4890	300	-290	160	230	-260	930
Trailing Edge Flap Disagree (1 =< Indicated Flap < 15)	VREF40+30	6330	410	-340	230	310	-300	1060
Trailing Edge Flap Disagree (15 =< Indicated Flap < 30)	VREF15	5430	340	-330	180	270	-280	990
Trailing Edge Flap Disagree (30 =< Indicated Flap < 40)	VREF30	4890	300	-290	160	230	-260	930
Trailing Edge Flaps Up Landing (Flaps Up)	VREF40+40	6890	430	-370	250	340	-320	1100

Actual (unfactored) distances are shown.

Includes a distance from runway threshold to touchdown associated with a flare time of 7 seconds.

Assumes maximum autobrake setting and maximum available reverse thrust.

For "One Engine Inoperative Landing - Flaps 30" autoland, increase the landing distance by 700 feet.

*For landing distance at or below 8000 ft pressure altitude, apply the STD altitude adjustment.

For altitudes higher than 8000 ft, first apply the STD altitude adjustment to 8000 ft,

then apply the HIGH altitude adjustment for the incremental altitude above 8000 ft.

Continued on next page

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

737-8 Non-Normal Configuration Landing Distances - Wet Runway - 2 Position Tailskid
LEAP-1B Engines

LANDING DISTANCES AND ADJUSTMENTS (FEET)								
QRH NON-NORMAL CHECKLIST	REFERENCE SPEED (VREF)	RUNWAY SLOPE ADJUSTMENT PER 1%		TEMPERATURE ADJUSTMENT PER		APPROACH SPEED ADJUSTMENT PER 5 KNOTS ABOVE V_{REF}	REVERSE THRUST ADJUSTMENT	
		DOWN HILL	UP HILL	10° C ABOVE ISA	10° C BELOW ISA		ONE REVERSER	NO REVERSER
One Engine Inoperative Landing (Flaps 15)	VREF15	120	-100	190	-180	310	N/A	310
One Engine Inoperative Landing (Flaps 30)	VREF30	100	-80	160	-160	290	N/A	220
SPOILERS (Flaps 15)	VREF15	230	-190	220	-220	430	650	1420
SPOILERS (Flaps 30)	VREF30	210	-170	200	-200	410	500	1060
SPOILERS (Flaps 40)	VREF40	210	-170	190	-190	420	470	970
Stabilizer Trim Inoperative (Flaps 15)	VREF15+10	110	-100	190	-190	290	350	700
Trailing Edge Flap Asymmetry (1 =< Flap Lever =< 15)	VREF40+30	120	-100	200	-200	290	390	810
Trailing Edge Flap Asymmetry (Flap Lever 15 or 25)	VREF15	100	-90	170	-170	290	280	570
Trailing Edge Flap Asymmetry (Flap Lever 30)	VREF30	80	-70	150	-150	270	200	400
Trailing Edge Flap Disagree (1 =< Indicated Flap < 15)	VREF40+30	120	-100	200	-200	290	390	810
Trailing Edge Flap Disagree (15 =< Indicated Flap < 30)	VREF15	100	-90	170	-170	290	280	570
Trailing Edge Flap Disagree (30 =< Indicated Flap < 40)	VREF30	80	-70	150	-150	270	200	400
Trailing Edge Flaps Up Landing (Flaps Up)	VREF40+40	130	-110	220	-220	300	460	950

Actual (unfactored) distances are shown.

Includes a distance from runway threshold to touchdown associated with a flare time of 7 seconds.

Assumes maximum autobrake setting and maximum available reverse thrust.

For "One Engine Inoperative Landing - Flaps 30" autoland, increase the landing distance by 700 feet.

Continued on next page

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

737-8 Non-Normal Configuration Landing Distances - Dry Runway - 2 Position Tailskid
LEAP-1B Engines

LANDING DISTANCES AND ADJUSTMENTS (METER)								
QRH NON-NORMAL CHECKLIST	REFERENCE SPEED (VREF)	REFERENCE DISTANCE FOR 65000 KGS	WEIGHT ADJUSTMENT PER 5000 KILOGRAMS		ALTITUDE ADJUSTMENT PER 1000 FT ABOVE SEA LEVEL *		RUNWAY WIND ADJUSTMENT PER 10 KNOTS	
			ABOVE 65000 KGS	BELOW 65000 KGS	STD	HIGH	HEAD WIND	TAIL WIND
Airspeed Unreliable (Flaps 15)	VREF15	1555	80	-90	40	55	-65	210
Airspeed Unreliable (Flaps 30)	VREF30	1430	75	-80	40	50	-60	205
Airspeed Unreliable (Flaps 40)	VREF40	1370	90	-80	40	50	-60	200
All Flaps Up Landing (Flaps UP)	VREF40+55	2060	125	-95	60	75	-75	235
ANTISKID INOPERATIVE (Flaps 15)	VREF15	1930	120	-125	60	80	-100	340
ANTISKID INOPERATIVE (Flaps 30)	VREF30	1750	105	-110	55	70	-95	330
ANTISKID INOPERATIVE (Flaps 40)	VREF40	1685	125	-105	55	70	-95	325
Jammed or Restricted Flight Controls (Flaps 15)	VREF15+10	1555	80	-90	40	55	-65	210
LEADING EDGE FLAPS (Flaps 15)	VREF15+15	1630	85	-90	45	55	-65	215
LOSS OF BOTH ENGINE DRIVEN GENERATORS (Flaps 15)	VREF15	2260	150	-155	75	105	-115	380
LOSS OF BOTH ENGINE DRIVEN GENERATORS (Flaps 30)	VREF30	2075	135	-140	70	90	-110	370
LOSS OF BOTH ENGINE DRIVEN GENERATORS (Flaps 40)	VREF40	2010	160	-135	70	90	-110	365
LOSS OF SYSTEM A (Flaps 15)	VREF15	1440	85	-85	40	50	-65	205
LOSS OF SYSTEM A (Flaps 30)	VREF30	1310	80	-80	35	45	-60	195
LOSS OF SYSTEM A (Flaps 40)	VREF40	1255	95	-75	35	45	-60	195
LOSS OF SYSTEM A AND SYSTEM B (Flaps 15)	VREF15	1835	115	-110	50	70	-80	265
LOSS OF SYSTEM B (Flaps 15)	VREF15	1450	90	-90	40	55	-65	220
MANUAL REVERSION (Flaps 15)	VREF15	1835	115	-110	50	70	-80	265

Actual (unfactored) distances are shown.

Includes a distance from runway threshold to touchdown associated with a flare time of 7 seconds.

Assumes maximum autobrake setting and maximum available reverse thrust.

Manual braking is assumed for ANTISKID INOPERATIVE, LOSS OF BOTH ENGINE DRIVEN GENERATORS, LOSS OF SYSTEM B, LOSS OF SYSTEM A AND SYSTEM B and MANUAL REVERSION since autobrakes are inoperative.

*For landing distance at or below 8000 ft pressure altitude, apply the STD altitude adjustment.

For altitudes higher than 8000 ft, first apply the STD altitude adjustment to 8000 ft, then apply the HIGH altitude adjustment for the incremental altitude above 8000 ft.

Continued on next page

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

737-8 Non-Normal Configuration Landing Distances - Dry Runway - 2 Position Tailskid
LEAP-1B Engines

LANDING DISTANCES AND ADJUSTMENTS (METER)								
QRH NON-NORMAL CHECKLIST	REFERENCE SPEED (VREF)	RUNWAY SLOPE ADJUSTMENT PER 1%		TEMPERATURE ADJUSTMENT PER		APPROACH SPEED ADJUSTMENT PER 5 KNOTS ABOVE V_{REF}	REVERSE THRUST ADJUSTMENT	
		DOWN HILL	UP HILL	10° C ABOVE ISA	10° C BELOW ISA		ONE REVERSER	NO REVERSER
Airspeed Unreliable (Flaps 15)	VREF15	0	0	40	-45	N/A	5	10
Airspeed Unreliable (Flaps 30)	VREF30	0	0	40	-40	N/A	0	10
Airspeed Unreliable (Flaps 40)	VREF40	5	0	35	-35	N/A	0	10
All Flaps Up Landing (Flaps UP)	VREF40+55	5	0	55	-55	85	5	15
ANTISKID INOPERATIVE (Flaps 15)	VREF15	60	-50	55	-55	80	150	340
ANTISKID INOPERATIVE (Flaps 30)	VREF30	55	-45	50	-50	75	120	265
ANTISKID INOPERATIVE (Flaps 40)	VREF40	55	-45	45	-50	75	110	235
Jammed or Restricted Flight Controls (Flaps 15)	VREF15+10	0	0	40	-45	75	5	10
LEADING EDGE FLAPS (Flaps 15)	VREF15+15	5	0	45	-45	80	5	10
LOSS OF BOTH ENGINE DRIVEN GENERATORS (Flaps 15)	VREF15	90	-75	65	-65	115	275	640
LOSS OF BOTH ENGINE DRIVEN GENERATORS (Flaps 30)	VREF30	85	-65	60	-60	110	225	515
LOSS OF BOTH ENGINE DRIVEN GENERATORS (Flaps 40)	VREF40	85	-70	60	-60	110	210	475
LOSS OF SYSTEM A (Flaps 15)	VREF15	10	-5	40	-40	75	25	40
LOSS OF SYSTEM A (Flaps 30)	VREF30	10	-5	35	-35	75	20	30
LOSS OF SYSTEM A (Flaps 40)	VREF40	10	-5	35	-35	75	20	30
LOSS OF SYSTEM A AND SYSTEM B (Flaps 15)	VREF15	50	-40	50	-50	110	15	90
LOSS OF SYSTEM B (Flaps 15)	VREF15	25	-20	40	-40	70	45	70
MANUAL REVERSION (Flaps 15)	VREF15	50	-40	50	-50	110	15	90

Actual (unfactored) distances are shown.

Includes a distance from runway threshold to touchdown associated with a flare time of 7 seconds.

Assumes maximum autobrake setting and maximum available reverse thrust.

Manual braking is assumed for ANTISKID INOPERATIVE, LOSS OF BOTH ENGINE DRIVEN GENERATORS, LOSS OF SYSTEM B, LOSS OF SYSTEM A AND SYSTEM B and MANUAL REVERSION since autobrakes are inoperative.

Continued on next page

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

737-8 Non-Normal Configuration Landing Distances - Dry Runway - 2 Position Tailskid
LEAP-1B Engines

LANDING DISTANCES AND ADJUSTMENTS (METER)								
QRH NON-NORMAL CHECKLIST	REFERENCE SPEED (VREF)	REFERENCE DISTANCE FOR 65000 KGS	WEIGHT ADJUSTMENT PER 5000 KILOGRAMS		ALTITUDE ADJUSTMENT PER 1000 FT ABOVE SEA LEVEL *		RUNWAY WIND ADJUSTMENT PER 10 KNOTS	
			ABOVE 65000 KGS	BELOW 65000 KGS	STD	HIGH	HEAD WIND	TAIL WIND
One Engine Inoperative Landing (Flaps 15)	VREF15	1410	80	-85	35	50	-60	200
One Engine Inoperative Landing (Flaps 30)	VREF30	1290	70	-75	35	45	-60	195
SPOILERS (Flaps 15)	VREF15	1455	90	-90	40	55	-65	205
SPOILERS (Flaps 30)	VREF30	1345	80	-85	40	50	-60	200
SPOILERS (Flaps 40)	VREF40	1305	100	-85	40	50	-60	200
Stabilizer Trim Inoperative (Flaps 15)	VREF15+10	1555	80	-90	40	55	-65	210
Trailing Edge Flap Asymmetry (1 =< Flap Lever =< 15)	VREF40+30	1660	100	-85	45	60	-65	215
Trailing Edge Flap Asymmetry (Flap Lever 15 or 25)	VREF15	1410	80	-85	35	50	-60	200
Trailing Edge Flap Asymmetry (Flap Lever 30)	VREF30	1290	70	-75	35	45	-60	195
Trailing Edge Flap Disagree (1 =< Indicated Flap < 15)	VREF40+30	1660	100	-85	45	60	-65	215
Trailing Edge Flap Disagree (15 =< Indicated Flap < 30)	VREF15	1410	80	-85	35	50	-60	200
Trailing Edge Flap Disagree (30 =< Indicated Flap < 40)	VREF30	1290	70	-75	35	45	-60	195
Trailing Edge Flaps Up Landing (Flaps Up)	VREF40+40	1815	110	-90	50	65	-70	225

Actual (unfactored) distances are shown.

Includes a distance from runway threshold to touchdown associated with a flare time of 7 seconds.

Assumes maximum autobrake setting and maximum available reverse thrust.

For "One Engine Inoperative Landing - Flaps 30" autoland, increase the landing distance by 210 meters.

*For landing distance at or below 8000 ft pressure altitude, apply the STD altitude adjustment.

For altitudes higher than 8000 ft, first apply the STD altitude adjustment to 8000 ft,

then apply the HIGH altitude adjustment for the incremental altitude above 8000 ft.

Continued on next page

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

737-8 Non-Normal Configuration Landing Distances - Dry Runway - 2 Position Tailskid
LEAP-1B Engines

LANDING DISTANCES AND ADJUSTMENTS (METER)								
QRH NON-NORMAL CHECKLIST	REFERENCE SPEED (VREF)	RUNWAY SLOPE ADJUSTMENT PER 1%		TEMPERATURE ADJUSTMENT PER		APPROACH SPEED ADJUSTMENT PER 5 KNOTS ABOVE V_{REF}	REVERSE THRUST ADJUSTMENT	
		DOWN HILL	UP HILL	10° C ABOVE ISA	10° C BELOW ISA		ONE REVERSER	NO REVERSER
One Engine Inoperative Landing (Flaps 15)	VREF15	5	0	40	-40	75	N/A	5
One Engine Inoperative Landing (Flaps 30)	VREF30	5	0	35	-35	70	N/A	5
SPOILERS (Flaps 15)	VREF15	15	-10	40	-40	85	50	85
SPOILERS (Flaps 30)	VREF30	15	-10	35	-35	85	45	70
SPOILERS (Flaps 40)	VREF40	15	-15	35	-35	90	45	70
Stabilizer Trim Inoperative (Flaps 15)	VREF15+10	0	0	40	-45	75	5	10
Trailing Edge Flap Asymmetry (1 =< Flap Lever =< 15)	VREF40+30	0	0	45	-45	80	5	10
Trailing Edge Flap Asymmetry (Flap Lever 15 or 25)	VREF15	0	0	40	-40	75	5	10
Trailing Edge Flap Asymmetry (Flap Lever 30)	VREF30	0	0	35	-35	70	0	10
Trailing Edge Flap Disagree (1 =< Indicated Flap < 15)	VREF40+30	0	0	45	-45	80	5	10
Trailing Edge Flap Disagree (15 =< Indicated Flap < 30)	VREF15	0	0	40	-40	75	5	10
Trailing Edge Flap Disagree (30 =< Indicated Flap < 40)	VREF30	0	0	35	-35	70	0	10
Trailing Edge Flaps Up Landing (Flaps Up)	VREF40+40	0	0	50	-50	80	5	10

Actual (unfactored) distances are shown.

Includes a distance from runway threshold to touchdown associated with a flare time of 7 seconds.

Assumes maximum autobrake setting and maximum available reverse thrust.

For "One Engine Inoperative Landing - Flaps 30" autoland, increase the landing distance by 210 meters.

Continued on next page

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

737-8 Non-Normal Configuration Landing Distances - Wet Runway - 2 Position Tailskid
LEAP-1B Engines

LANDING DISTANCES AND ADJUSTMENTS (METER)								
QRH NON-NORMAL CHECKLIST	REFERENCE SPEED (VREF)	REFERENCE DISTANCE FOR 65000 KGS	WEIGHT ADJUSTMENT PER 5000 KILOGRAMS		ALTITUDE ADJUSTMENT PER 1000 FT ABOVE SEA LEVEL *		RUNWAY WIND ADJUSTMENT PER 10 KNOTS	
			ABOVE 65000 KGS	BELOW 65000 KGS	STD	HIGH	HEAD WIND	TAIL WIND
Airspeed Unreliable (Flaps 15)	VREF15	1760	125	-115	65	90	-90	315
Airspeed Unreliable (Flaps 30)	VREF30	1595	105	-105	55	75	-85	295
Airspeed Unreliable (Flaps 40)	VREF40	1535	125	-100	55	75	-80	290
All Flaps Up Landing (Flaps UP)	VREF40+55	2340	160	-140	90	120	-105	365
ANTISKID INOPERATIVE (Flaps 15)	VREF15	2245	175	-165	85	120	-130	480
ANTISKID INOPERATIVE (Flaps 30)	VREF30	2005	150	-140	75	105	-125	455
ANTISKID INOPERATIVE (Flaps 40)	VREF40	1910	170	-135	75	100	-120	445
Jammed or Restricted Flight Controls (Flaps 15)	VREF15+10	1760	125	-115	65	90	-90	315
LEADING EDGE FLAPS (Flaps 15)	VREF15+15	1885	130	-125	70	95	-95	325
LOSS OF BOTH ENGINE DRIVEN GENERATORS (Flaps 15)	VREF15	2655	220	-200	110	160	-150	545
LOSS OF BOTH ENGINE DRIVEN GENERATORS (Flaps 30)	VREF30	2395	195	-180	95	135	-140	515
LOSS OF BOTH ENGINE DRIVEN GENERATORS (Flaps 40)	VREF40	2290	220	-170	95	130	-140	505
LOSS OF SYSTEM A (Flaps 15)	VREF15	1810	145	-135	70	105	-100	350
LOSS OF SYSTEM A (Flaps 30)	VREF30	1590	120	-115	60	85	-90	325
LOSS OF SYSTEM A (Flaps 40)	VREF40	1515	145	-110	60	80	-90	315
LOSS OF SYSTEM A AND SYSTEM B (Flaps 15)	VREF15	2465	190	-175	95	140	-135	465
LOSS OF SYSTEM B (Flaps 15)	VREF15	1830	140	-130	70	100	-100	360
MANUAL REVERSION (Flaps 15)	VREF15	2465	190	-175	95	140	-135	465

Actual (unfactored) distances are shown.

Includes a distance from runway threshold to touchdown associated with a flare time of 7 seconds.

Assumes maximum autobrake setting and maximum available reverse thrust.

Manual braking is assumed for ANTISKID INOPERATIVE, LOSS OF BOTH ENGINE DRIVEN GENERATORS, LOSS OF SYSTEM B, LOSS OF SYSTEM A AND SYSTEM B and MANUAL REVERSION since autobrakes are inoperative.

*For landing distance at or below 8000 ft pressure altitude, apply the STD altitude adjustment.

For altitudes higher than 8000 ft, first apply the STD altitude adjustment to 8000 ft, then apply the HIGH altitude adjustment for the incremental altitude above 8000 ft.

Continued on next page

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

737-8 Non-Normal Configuration Landing Distances - Wet Runway - 2 Position Tailskid
LEAP-1B Engines

LANDING DISTANCES AND ADJUSTMENTS (METER)								
QRH NON-NORMAL CHECKLIST	REFERENCE SPEED (VREF)	RUNWAY SLOPE ADJUSTMENT PER 1%		TEMPERATURE ADJUSTMENT PER		APPROACH SPEED ADJUSTMENT PER 5 KNOTS ABOVE V_{REF}	REVERSE THRUST ADJUSTMENT	
		DOWN HILL	UP HILL	10° C ABOVE ISA	10° C BELOW ISA		ONE REVERSER	NO REVERSER
Airspeed Unreliable (Flaps 15)	VREF15	35	-30	60	-60	N/A	105	215
Airspeed Unreliable (Flaps 30)	VREF30	30	-25	50	-50	N/A	80	155
Airspeed Unreliable (Flaps 40)	VREF40	30	-25	50	-50	N/A	70	135
All Flaps Up Landing (Flaps UP)	VREF40+55	50	-40	80	-80	95	195	400
ANTISKID INOPERATIVE (Flaps 15)	VREF15	95	-75	75	-75	100	295	745
ANTISKID INOPERATIVE (Flaps 30)	VREF30	80	-65	65	-65	95	215	530
ANTISKID INOPERATIVE (Flaps 40)	VREF40	80	-65	60	-60	95	190	460
Jammed or Restricted Flight Controls (Flaps 15)	VREF15+10	35	-30	60	-60	90	105	215
LEADING EDGE FLAPS (Flaps 15)	VREF15+15	40	-35	65	-60	95	125	260
LOSS OF BOTH ENGINE DRIVEN GENERATORS (Flaps 15)	VREF15	145	-110	90	-90	135	535	1540
LOSS OF BOTH ENGINE DRIVEN GENERATORS (Flaps 30)	VREF30	130	-100	80	-80	130	410	1115
LOSS OF BOTH ENGINE DRIVEN GENERATORS (Flaps 40)	VREF40	125	-95	75	-75	125	365	960
LOSS OF SYSTEM A (Flaps 15)	VREF15	60	-50	65	-65	115	150	275
LOSS OF SYSTEM A (Flaps 30)	VREF30	50	-40	55	-55	105	100	175
LOSS OF SYSTEM A (Flaps 40)	VREF40	45	-40	50	-50	105	85	145
LOSS OF SYSTEM A AND SYSTEM B (Flaps 15)	VREF15	115	-95	85	-80	155	130	560
LOSS OF SYSTEM B (Flaps 15)	VREF15	50	-45	60	-60	100	150	290
MANUAL REVERSION (Flaps 15)	VREF15	115	-95	85	-80	155	130	560

Actual (unfactored) distances are shown.

Includes a distance from runway threshold to touchdown associated with a flare time of 7 seconds.

Assumes maximum autobrake setting and maximum available reverse thrust.

Manual braking is assumed for ANTISKID INOPERATIVE, LOSS OF BOTH ENGINE DRIVEN GENERATORS, LOSS OF SYSTEM B, LOSS OF SYSTEM A AND SYSTEM B and MANUAL REVERSION since autobrakes are inoperative.

Continued on next page

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

737-8 Non-Normal Configuration Landing Distances - Wet Runway - 2 Position Tailskid
LEAP-1B Engines

LANDING DISTANCES AND ADJUSTMENTS (METER)								
QRH NON-NORMAL CHECKLIST	REFERENCE SPEED (VREF)	REFERENCE DISTANCE FOR 65000 KGS	WEIGHT ADJUSTMENT PER 5000 KILOGRAMS		ALTITUDE ADJUSTMENT PER 1000 FT ABOVE SEA LEVEL *		RUNWAY WIND ADJUSTMENT PER 10 KNOTS	
			ABOVE 65000 KGS	BELOW 65000 KGS	STD	HIGH	HEAD WIND	TAIL WIND
One Engine Inoperative Landing (Flaps 15)	VREF15	1645	125	-115	60	85	-90	325
One Engine Inoperative Landing (Flaps 30)	VREF30	1470	105	-100	50	70	-85	300
SPOILERS (Flaps 15)	VREF15	1880	155	-140	75	115	-105	365
SPOILERS (Flaps 30)	VREF30	1705	135	-125	70	100	-100	345
SPOILERS (Flaps 40)	VREF40	1650	165	-125	70	95	-95	340
Stabilizer Trim Inoperative (Flaps 15)	VREF15+10	1760	125	-115	65	90	-90	315
Trailing Edge Flap Asymmetry (1 =< Flap Lever =< 15)	VREF40+30	1860	140	-115	70	95	-90	325
Trailing Edge Flap Asymmetry (Flap Lever 15 or 25)	VREF15	1590	115	-110	55	80	-85	300
Trailing Edge Flap Asymmetry (Flap Lever 30)	VREF30	1435	100	-95	50	70	-80	280
Trailing Edge Flap Disagree (1 =< Indicated Flap < 15)	VREF40+30	1860	140	-115	70	95	-90	325
Trailing Edge Flap Disagree (15 =< Indicated Flap < 30)	VREF15	1590	115	-110	55	80	-85	300
Trailing Edge Flap Disagree (30 =< Indicated Flap < 40)	VREF30	1435	100	-95	50	70	-80	280
Trailing Edge Flaps Up Landing (Flaps Up)	VREF40+40	2025	145	-125	75	105	-95	335

Actual (unfactored) distances are shown.

Includes a distance from runway threshold to touchdown associated with a flare time of 7 seconds.

Assumes maximum autobrake setting and maximum available reverse thrust.

For "One Engine Inoperative Landing - Flaps 30" autoland, increase the landing distance by 210 meters.

*For landing distance at or below 8000 ft pressure altitude, apply the STD altitude adjustment.

For altitudes higher than 8000 ft, first apply the STD altitude adjustment to 8000 ft,

then apply the HIGH altitude adjustment for the incremental altitude above 8000 ft.

Continued on next page

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

737-8 Non-Normal Configuration Landing Distances - Wet Runway - 2 Position Tailskid
LEAP-1B Engines

LANDING DISTANCES AND ADJUSTMENTS (METER)								
QRH NON-NORMAL CHECKLIST	REFERENCE SPEED (VREF)	RUNWAY SLOPE ADJUSTMENT PER 1%		TEMPERATURE ADJUSTMENT PER		APPROACH SPEED ADJUSTMENT PER 5 KNOTS ABOVE V_{REF}	REVERSE THRUST ADJUSTMENT	
		DOWN HILL	UP HILL	10° C ABOVE ISA	10° C BELOW ISA		ONE REVERSER	NO REVERSER
One Engine Inoperative Landing (Flaps 15)	VREF15	35	-30	55	-55	95	N/A	95
One Engine Inoperative Landing (Flaps 30)	VREF30	30	-25	50	-50	90	N/A	65
SPOILERS (Flaps 15)	VREF15	70	-60	65	-65	130	200	435
SPOILERS (Flaps 30)	VREF30	65	-50	60	-60	125	150	325
SPOILERS (Flaps 40)	VREF40	65	-55	55	-60	130	140	295
Stabilizer Trim Inoperative (Flaps 15)	VREF15+10	35	-30	60	-60	90	105	215
Trailing Edge Flap Asymmetry (1 =< Flap Lever =< 15)	VREF40+30	35	-30	60	-60	90	120	245
Trailing Edge Flap Asymmetry (Flap Lever 15 or 25)	VREF15	30	-25	50	-50	90	85	175
Trailing Edge Flap Asymmetry (Flap Lever 30)	VREF30	25	-25	45	-45	85	60	125
Trailing Edge Flap Disagree (1 =< Indicated Flap < 15)	VREF40+30	35	-30	60	-60	90	120	245
Trailing Edge Flap Disagree (15 =< Indicated Flap < 30)	VREF15	30	-25	50	-50	90	85	175
Trailing Edge Flap Disagree (30 =< Indicated Flap < 40)	VREF30	25	-25	45	-45	85	60	125
Trailing Edge Flaps Up Landing (Flaps Up)	VREF40+40	40	-35	70	-70	90	140	290

Actual (unfactored) distances are shown.

Includes a distance from runway threshold to touchdown associated with a flare time of 7 seconds.

Assumes maximum autobrake setting and maximum available reverse thrust.

For "One Engine Inoperative Landing - Flaps 30" autoland, increase the landing distance by 210 meters.

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

Performance Configuration A180522

The airplane configuration associated with the performance data in AFM-DPI is shown below.

	THRUST	FLAPS	GEAR
TAKEOFF	Takeoff on all operating engines.	25, 15, 10, 5 or 1	Down
1ST SEGMENT CLIMB	Takeoff on operating engine.	25, 15, 10, 5 or 1	Down
2ND SEGMENT CLIMB	Same as 1st segment.	25, 15, 10, 5 or 1	Up
3RD SEGMENT	Same as 1st segment.	25, 15, 10, 5 or 1	Up
FINAL TAKEOFF CLIMB	Maximum continuous on operating engine.	Up	Up
ENROUTE CLIMB	Maximum continuous on operating engine.	Up	Up
APPROACH CLIMB	Go-around on operating engine.	15, 5*, or 1**	Up
LANDING CLIMB	Maximum available in 8 seconds on all engines.	40, 30, or 15**	Down
LANDING	Idle on all engines at touchdown.	40 or 30	Down

* For operation with Flap 5 Approach Climb settings see AFM-DPI Authorized Alternate Performance Capability in the basic AFM.

** Only for use when calculating takeoff performance limits due to fuel jettison considerations, not as a normal landing configuration.

Alternate performance capabilities are noted in the Alternate Performance Capability section of this AFM appendix. Application of valid alternate performance capabilities is controlled by use of the airplane serial number in the Authorized Alternate Performance Capability in the basic AFM. Use of the alternate performance capability is only valid when it is consistent with the current airplane configuration.

The use of a valid alternate performance capability requires the operator to adhere to the limitations and operational considerations stipulated in the associated appendix where provided.

Use of the ten minute takeoff alternate performance, where authorized, is limited to situations where an engine failure actually occurs and an obstacle is in the takeoff flight path.

Use of the Alternate Center of Gravity performance, where authorized, requires alternate forward center of gravities be provided on the Center of Gravity Chart in section of the basic AFM. In addition, the airplane must be loaded to meet the alternate forward center of gravities reflected on that page.

Takeoff performance calculated using AFM-DPI for ECS on is based on a normal ECS configuration (2 A/C packs on).

Continued on next page

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

Scheduled landing performance assumes that the automatic speed brake system is operative, except when manual speed brakes are selected.

Performance is applicable to APU on and off.

Takeoff, Enroute, and Landing performance for EEC's in the alternate mode is provided in AFM-DPI as an alternate performance by selecting ALTERNATE EEC MODE.

Tires:

Approved tires with a speed rating of at least 235 mph and a one-time overspeed capability of 260 mph must be installed on the airplane to utilize the allowable tire speed limit of 232 mph in AFM-DPI.

An option exists in AFM-DPI to utilize a reduced tire speed limit of 225 mph for approved tires with a speed rating of at least 225 mph and a one-time overspeed capability of 250 mph.

Takeoff and Landing on Wet Skid-Resistant Surfaces:

Takeoff and landing performance on wet skid-resistant surfaces, such as grooved runways or porous friction course runways, is provided in AFM-DPI by selecting WET SK-R. This data may be used only if the runway is constructed and maintained to meet the Friction Level Classification for Runway Pavement Surface defined in AC 150/5320-12C dated 3/18/97 or its equivalent. Operational approval of the wet skid resistant data must be obtained from the appropriate regulatory authority.

The improved landing performance on wet skid-resistant surfaces may not be applicable to all airplane serial numbers covered by the basic AFM. Refer to the Authorized Alternate Performance Capability of the basic AFM to determine airplane serial number and configuration applicability.

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

Recommended Takeoff Stabilizer Setting A191005

The Flight Management Computer (FMC) provides Recommended Takeoff Stabilizer Settings corrected for reduced takeoff thrust, resulting from any combination of selected "pushbutton" derate and/or the Assumed Temperature Thrust Reduction Method.

737-8

- LEAP-1B27 ENGINES AT FULL THRUST AND UP TO, BUT NOT INCLUDING, 10% DERATE.*
- OBSERVE CENTER OF GRAVITY LIMITS.
- USE LINEAR INTERPOLATION AND ROUND TO THE NEAREST 1/4 UNIT.

FLAPS 1, 5

CENTER OF GRAVITY - PERCENT MAC	TAKEOFF GROSS WEIGHT - 1000 LB.						
	80	100	120	140	160	180	185
	STABILIZER TRIM UNITS						
6	4 3/4	6 1/4	7 1/4	7 3/4	8 1/2	8 1/2	8 1/2
8	4 1/2	5 3/4	6 3/4	7 1/2	8	8 1/2	8 1/2
10	4 1/4	5 1/2	6 1/2	7 1/4	7 3/4	8 1/4	8 1/4
12	4 1/4	5 1/4	6 1/4	7	7 1/2	7 3/4	7 3/4
14	4	5	6	6 1/2	7	7 1/2	7 1/2
16	3 3/4	4 3/4	5 3/4	6 1/4	6 3/4	7 1/4	7 1/4
18	3 1/2	4 1/2	5 1/4	6	6 1/2	6 3/4	6 3/4
20	3 1/4	4 1/4	5	5 3/4	6	6 1/2	6 1/2
22	3	3 3/4	4 3/4	5 1/4	5 3/4	6	6
24	3	3 1/2	4 1/4	5	5 1/4	5 3/4	5 3/4
26	3	3 1/4	4	4 1/2	5	5 1/4	5 1/4
28	3	3	3 3/4	4 1/4	4 1/2	5	5
30	3	3	3 1/4	3 3/4	4 1/4	4 1/2	4 1/2
32	3	3	3	3 1/2	3 3/4	4 1/4	4 1/4
34	3	3	3	3	3 1/2	3 3/4	3 3/4
36	3	3	3	3	3	3 1/2	3 1/2

FLAPS 10, 15, 25

CENTER OF GRAVITY - PERCENT MAC	TAKEOFF GROSS WEIGHT - 1000 LB.						
	80	100	120	140	160	180	185
	STABILIZER TRIM UNITS						
6	4	5 1/4	7	7 3/4	8 1/2	8 1/2	8 1/2
8	3 3/4	5	6 1/2	7 1/2	8	8 1/2	8 1/2
10	3 1/2	4 1/2	6	7	7 1/2	8	8
12	3 1/4	4 1/4	5 1/2	6 1/2	7	7 1/2	7 1/2
14	3	4	5 1/4	6	6 1/2	7	7
16	3	3 3/4	4 3/4	5 1/2	6	6 1/2	6 1/2
18	3	3 1/4	4 1/4	5	5 1/2	6	6
20	3	3	4	4 3/4	5 1/4	5 1/2	5 1/2
22	3	3	3 1/2	4 1/4	4 3/4	5 1/4	5 1/4
24	3	3	3 1/4	3 3/4	4 1/4	4 3/4	4 3/4
26	3	3	3	3 1/2	4	4 1/4	4 1/4
28	3	3	3	3	3 1/2	3 3/4	3 3/4
30	3	3	3	3	3	3 1/2	3 1/2
32	3	3	3	3	3	3	3
34	3	3	3	3	3	3	3
36	3	3	3	3	3	3	3

AFMSTA801_REV_M_A3004048H1_D.RJENS_11-19-19

* DERATE RESULTING FROM ANY COMBINATION OF SELECTED PUSHBUTTON DERATE AND/OR ASSUMED TEMPERATURE THRUST REDUCTION

Continued on next page

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

Section Recommended Takeoff Stabilizer Setting, continued from previous page A191005

737-8

- LEAP-1B27 ENGINES AT FULL THRUST AND UP TO, BUT NOT INCLUDING, 10% DERATE.*
- OBSERVE CENTER OF GRAVITY LIMITS.
- USE LINEAR INTERPOLATION AND ROUND TO THE NEAREST 1/4 UNIT.

FLAPS 1, 5

CENTER OF GRAVITY - PERCENT MAC	TAKEOFF GROSS WEIGHT - 1000 KG.						
	40	50	60	70	80	82	84
	STABILIZER TRIM UNITS						
6	5 1/4	6 3/4	7 1/2	8 1/4	8 1/2	8 1/2	8 1/2
8	5	6 1/4	7 1/4	8	8 1/2	8 1/2	8 1/2
10	4 3/4	6	7	7 1/2	8	8 1/4	8 1/4
12	4 1/2	5 3/4	6 3/4	7 1/4	7 3/4	7 3/4	7 3/4
14	4 1/4	5 1/2	6 1/4	7	7 1/2	7 1/2	7 1/2
16	4	5 1/4	6	6 3/4	7	7 1/4	7 1/4
18	3 3/4	5	5 3/4	6 1/4	6 3/4	6 3/4	6 3/4
20	3 1/2	4 1/2	5 1/2	6	6 1/2	6 1/2	6 1/2
22	3 1/4	4 1/4	5	5 1/2	6	6 1/4	6 1/4
24	3	4	4 3/4	5 1/4	5 3/4	5 3/4	5 3/4
26	3	3 1/2	4 1/4	4 3/4	5 1/4	5 1/2	5 1/2
28	3	3 1/4	4	4 1/2	5	5	5
30	3	3	3 3/4	4	4 1/2	4 3/4	4 3/4
32	3	3	3 1/4	3 3/4	4 1/4	4 1/4	4 1/4
34	3	3	3	3 1/2	3 3/4	4	4
36	3	3	3	3	3 1/2	3 1/2	3 1/2

FLAPS 10, 15, 25

CENTER OF GRAVITY - PERCENT MAC	TAKEOFF GROSS WEIGHT - 1000 KG.						
	40	50	60	70	80	82	84
	STABILIZER TRIM UNITS						
6	4 1/2	6	7 1/2	8 1/2	8 1/2	8 1/2	8 1/2
8	4 1/4	5 3/4	7	7 3/4	8 1/2	8 1/2	8 1/2
10	4	5 1/4	6 1/2	7 1/4	7 3/4	8	8
12	3 3/4	5	6	6 3/4	7 1/4	7 1/2	7 1/2
14	3 1/2	4 1/2	5 3/4	6 1/4	6 3/4	7	7
16	3 1/4	4 1/4	5 1/4	5 3/4	6 1/4	6 1/2	6 1/2
18	3	3 3/4	4 3/4	5 1/2	6	6	6
20	3	3 1/2	4 1/2	5	5 1/2	5 1/2	5 1/2
22	3	3 1/4	4	4 1/2	5	5 1/4	5 1/4
24	3	3	3 1/2	4 1/4	4 3/4	4 3/4	4 3/4
26	3	3	3 1/4	3 3/4	4 1/4	4 1/4	4 1/4
28	3	3	3	3 1/4	3 3/4	4	4
30	3	3	3	3	3 1/4	3 1/2	3 1/2
32	3	3	3	3	3	3	3
34	3	3	3	3	3	3	3
36	3	3	3	3	3	3	3

AFMSTA801_REV_M_A300404BH3_D.R.JENS_11-19-19

* DERATE RESULTING FROM ANY COMBINATION OF SELECTED PUSHBUTTON DERATE AND/OR ASSUMED TEMPERATURE THRUST REDUCTION

Continued on next page

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

737-8

- LEAP-1B27 ENGINES AT 10% AND UP TO, BUT NOT INCLUDING, 20% DERATE.*
- OBSERVE CENTER OF GRAVITY LIMITS.
- USE LINEAR INTERPOLATION AND ROUND TO THE NEAREST 1/4 UNIT.

FLAPS 1, 5

CENTER OF GRAVITY - PERCENT MAC	TAKEOFF GROSS WEIGHT - 1000 LB.						
	80	100	120	140	160	180	185
	STABILIZER TRIM UNITS						
6	6	7 1/4	8	8 1/2	8 1/2	8 1/2	8 1/2
8	5 3/4	6 3/4	7 3/4	8 1/4	8 1/2	8 1/2	8 1/2
10	5 1/4	6 1/2	7 1/4	7 3/4	8 1/4	8 1/2	8 1/2
12	5	6 1/4	7	7 1/2	8	8 1/4	8 1/4
14	4 3/4	6	6 3/4	7 1/4	7 1/2	7 3/4	7 3/4
16	4 1/2	5 1/2	6 1/4	6 3/4	7 1/4	7 1/2	7 1/2
18	4 1/4	5 1/4	6	6 1/2	7	7 1/4	7 1/4
20	4	5	5 3/4	6 1/4	6 1/2	6 3/4	6 3/4
22	3 3/4	4 1/2	5 1/4	5 3/4	6 1/4	6 1/2	6 1/2
24	3 1/2	4 1/4	5	5 1/2	5 3/4	6	6
26	3	4	4 3/4	5	5 1/2	5 3/4	5 3/4
28	3	3 1/2	4 1/4	4 3/4	5	5 1/4	5 1/4
30	3	3 1/4	4	4 1/2	4 3/4	5	5
32	3	3	3 1/2	4	4 1/4	4 3/4	4 3/4
34	3	3	3 1/4	3 3/4	4	4 1/4	4 1/4
36	3	3	3	3 1/4	3 1/2	4	4

FLAPS 10, 15, 25

CENTER OF GRAVITY - PERCENT MAC	TAKEOFF GROSS WEIGHT - 1000 LB.						
	80	100	120	140	160	180	185
	STABILIZER TRIM UNITS						
6	5	6 1/4	7 3/4	8 1/2	8 1/2	8 1/2	8 1/2
8	4 3/4	6	7 1/4	8	8 1/2	8 1/2	8 1/2
10	4 1/2	5 3/4	6 3/4	7 1/2	8	8 1/4	8 1/4
12	4 1/4	5 1/4	6 1/2	7	7 1/2	7 3/4	7 3/4
14	3 3/4	5	6	6 1/2	7	7 1/4	7 1/4
16	3 1/2	4 1/2	5 1/2	6	6 1/2	6 3/4	6 3/4
18	3 1/4	4 1/4	5	5 3/4	6	6 1/2	6 1/2
20	3	3 3/4	4 3/4	5 1/4	5 3/4	6	6
22	3	3 1/2	4 1/4	4 3/4	5 1/4	5 1/2	5 1/2
24	3	3	3 3/4	4 1/2	4 3/4	5 1/4	5 1/4
26	3	3	3 1/2	4	4 1/2	4 3/4	4 3/4
28	3	3	3	3 1/2	4	4 1/4	4 1/4
30	3	3	3	3 1/4	3 1/2	3 3/4	3 3/4
32	3	3	3	3	3	3 1/2	3 1/2
34	3	3	3	3	3	3	3
36	3	3	3	3	3	3	3

AFMSTA801_REV_M_A300404BM1_D.RJENS_11-19-19

* DERATE RESULTING FROM ANY COMBINATION OF SELECTED PUSHBUTTON DERATE AND/OR ASSUMED TEMPERATURE THRUST REDUCTION

Continued on next page

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

737-8

- LEAP-1B27 ENGINES AT 10% AND UP TO, BUT NOT INCLUDING, 20% DERATE.*
- OBSERVE CENTER OF GRAVITY LIMITS.
- USE LINEAR INTERPOLATION AND ROUND TO THE NEAREST 1/4 UNIT.

FLAPS 1, 5

CENTER OF GRAVITY - PERCENT MAC	TAKEOFF GROSS WEIGHT - 1000 KG.						
	40	50	60	70	80	82	84
	STABILIZER TRIM UNITS						
6	6 1/2	7 1/2	8 1/4	8 1/2	8 1/2	8 1/2	8 1/2
8	6 1/4	7 1/4	8	8 1/2	8 1/2	8 1/2	8 1/2
10	5 3/4	7	7 3/4	8 1/4	8 1/2	8 1/2	8 1/2
12	5 1/2	6 3/4	7 1/4	7 3/4	8 1/4	8 1/4	8 1/4
14	5 1/4	6 1/4	7	7 1/2	7 3/4	7 3/4	7 3/4
16	5	6	6 3/4	7 1/4	7 1/2	7 1/2	7 1/2
18	4 3/4	5 3/4	6 1/4	6 3/4	7	7 1/4	7 1/4
20	4 1/4	5 1/4	6	6 1/2	6 3/4	6 3/4	6 3/4
22	4	5	5 3/4	6	6 1/2	6 1/2	6 1/2
24	3 3/4	4 3/4	5 1/4	5 3/4	6	6	6
26	3 1/2	4 1/4	5	5 1/4	5 3/4	5 3/4	5 3/4
28	3 1/4	4	4 1/2	5	5 1/4	5 1/2	5 1/2
30	3	3 1/2	4 1/4	4 1/2	5	5	5
32	3	3 1/4	3 3/4	4 1/4	4 1/2	4 3/4	4 3/4
34	3	3	3 1/2	3 3/4	4 1/4	4 1/4	4 1/4
36	3	3	3 1/4	3 1/2	3 3/4	4	4

FLAPS 10, 15, 25

CENTER OF GRAVITY - PERCENT MAC	TAKEOFF GROSS WEIGHT - 1000 KG.						
	40	50	60	70	80	82	84
	STABILIZER TRIM UNITS						
6	5 1/2	7	8 1/4	8 1/2	8 1/2	8 1/2	8 1/2
8	5 1/4	6 3/4	7 3/4	8 1/4	8 1/2	8 1/2	8 1/2
10	5	6 1/4	7 1/4	7 3/4	8 1/4	8 1/4	8 1/4
12	4 1/2	5 3/4	6 3/4	7 1/4	7 3/4	7 3/4	7 3/4
14	4 1/4	5 1/2	6 1/4	6 3/4	7 1/4	7 1/4	7 1/4
16	4	5	5 3/4	6 1/4	6 3/4	6 3/4	6 3/4
18	3 1/2	4 3/4	5 1/2	6	6 1/4	6 1/2	6 1/2
20	3 1/4	4 1/4	5	5 1/2	6	6	6
22	3 1/4	4	4 3/4	5 1/4	5 1/2	5 1/2	5 1/2
24	3	3 1/2	4 1/4	4 3/4	5	5 1/4	5 1/4
26	3	3 1/4	3 3/4	4 1/4	4 3/4	4 3/4	4 3/4
28	3	3	3 1/2	3 3/4	4 1/4	4 1/4	4 1/4
30	3	3	3	3 1/2	3 3/4	3 3/4	3 3/4
32	3	3	3	3	3 1/4	3 1/2	3 1/2
34	3	3	3	3	3	3	3
36	3	3	3	3	3	3	3

AFMSTA801_REV_M_A300404BM3_D.RJENS_11-19-19

* DERATE RESULTING FROM ANY COMBINATION OF SELECTED PUSHBUTTON DERATE AND/OR ASSUMED TEMPERATURE THRUST REDUCTION

Continued on next page

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

Section Recommended Takeoff Stabilizer Setting, continued from previous page A191005

737-8

- LEAP-1B27 ENGINES AT 20% AND GREATER DERATE.*
- OBSERVE CENTER OF GRAVITY LIMITS.
- USE LINEAR INTERPOLATION AND ROUND TO THE NEAREST 1/4 UNIT.

FLAPS 1, 5

CENTER OF GRAVITY - PERCENT MAC	TAKEOFF GROSS WEIGHT - 1000 LB.						
	80	100	120	140	160	180	185
	STABILIZER TRIM UNITS						
6	7	8 1/4	8 1/2	8 1/2	8 1/2	8 1/2	8 1/2
8	6 3/4	7 3/4	8 1/2	8 1/2	8 1/2	8 1/2	8 1/2
10	6 1/2	7 1/2	8	8 1/2	8 1/2	8 1/2	8 1/2
12	6 1/4	7 1/4	7 3/4	8 1/4	8 1/2	8 1/2	8 1/2
14	5 3/4	6 3/4	7 1/2	7 3/4	8	8 1/4	8 1/4
16	5 1/2	6 1/2	7	7 1/2	7 3/4	8	8
18	5 1/4	6 1/4	6 3/4	7	7 1/4	7 1/2	7 1/2
20	5	5 3/4	6 1/2	6 3/4	7	7 1/4	7 1/4
22	4 1/2	5 1/2	6	6 1/2	6 3/4	6 3/4	6 3/4
24	4 1/4	5	5 3/4	6	6 1/4	6 1/2	6 1/2
26	4	4 3/4	5 1/4	5 3/4	6	6	6
28	3 1/2	4 1/4	5	5 1/4	5 1/2	5 3/4	5 3/4
30	3 1/4	4	4 1/2	5	5 1/4	5 1/2	5 1/2
32	3	3 1/2	4 1/4	4 1/2	4 3/4	5	5
34	3	3 1/4	3 3/4	4 1/4	4 1/2	4 3/4	4 3/4
36	3	3	3 1/2	3 3/4	4	4 1/4	4 1/4

FLAPS 10, 15, 25

CENTER OF GRAVITY - PERCENT MAC	TAKEOFF GROSS WEIGHT - 1000 LB.						
	80	100	120	140	160	180	185
	STABILIZER TRIM UNITS						
6	6 1/4	7 1/2	8 1/2	8 1/2	8 1/2	8 1/2	8 1/2
8	6	7	8	8 1/2	8 1/2	8 1/2	8 1/2
10	5 1/2	6 3/4	7 3/4	8 1/4	8 1/2	8 1/2	8 1/2
12	5 1/4	6 1/4	7 1/4	7 3/4	8	8 1/4	8 1/4
14	5	6	6 3/4	7 1/4	7 1/2	7 3/4	7 3/4
16	4 1/2	5 1/2	6 1/4	6 3/4	7	7 1/4	7 1/4
18	4 1/4	5 1/4	5 3/4	6 1/4	6 1/2	6 3/4	6 3/4
20	3 3/4	4 3/4	5 1/2	6	6 1/4	6 1/2	6 1/2
22	3 1/4	4 1/4	5	5 1/2	5 3/4	6	6
24	3	4	4 1/2	5	5 1/4	5 1/2	5 1/2
26	3	3 1/2	4 1/4	4 1/2	5	5 1/4	5 1/4
28	3	3	3 3/4	4 1/4	4 1/2	4 3/4	4 3/4
30	3	3	3 1/4	3 3/4	4	4 1/4	4 1/4
32	3	3	3	3 1/4	3 1/2	3 3/4	3 3/4
34	3	3	3	3	3 1/4	3 1/2	3 1/2
36	3	3	3	3	3	3	3

AFMSTA801_REV_M_A300404BL1_D.RJENS_11-19-19

* DERATE RESULTING FROM ANY COMBINATION OF SELECTED PUSHBUTTON DERATE AND/OR ASSUMED TEMPERATURE THRUST REDUCTION

Continued on next page

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

Section Recommended Takeoff Stabilizer Setting, continued from previous page A191005

737-8

- LEAP-1B27 ENGINES AT 20% AND GREATER DERATE.*
- OBSERVE CENTER OF GRAVITY LIMITS.
- USE LINEAR INTERPOLATION AND ROUND TO THE NEAREST 1/4 UNIT.

FLAPS 1, 5

CENTER OF GRAVITY - PERCENT MAC	TAKEOFF GROSS WEIGHT - 1000 KG.						
	40	50	60	70	80	82	84
	STABILIZER TRIM UNITS						
6	7 1/2	8 1/2	8 1/2	8 1/2	8 1/2	8 1/2	8 1/2
8	7 1/4	8 1/4	8 1/2	8 1/2	8 1/2	8 1/2	8 1/2
10	7	7 3/4	8 1/4	8 1/2	8 1/2	8 1/2	8 1/2
12	6 1/2	7 1/2	8	8 1/4	8 1/2	8 1/2	8 1/2
14	6 1/4	7 1/4	7 3/4	8	8 1/4	8 1/4	8 1/4
16	6	6 3/4	7 1/4	7 3/4	7 3/4	8	8
18	5 1/2	6 1/2	7	7 1/4	7 1/2	7 1/2	7 1/2
20	5 1/4	6 1/4	6 3/4	7	7 1/4	7 1/4	7 1/4
22	5	5 3/4	6 1/4	6 1/2	6 3/4	6 3/4	6 3/4
24	4 1/2	5 1/2	6	6 1/4	6 1/2	6 1/2	6 1/2
26	4 1/4	5	5 1/2	5 3/4	6	6	6
28	4	4 3/4	5 1/4	5 1/2	5 3/4	5 3/4	5 3/4
30	3 1/2	4 1/4	4 3/4	5	5 1/4	5 1/2	5 1/2
32	3 1/4	4	4 1/2	4 3/4	5	5	5
34	3	3 1/2	4	4 1/4	4 1/2	4 3/4	4 3/4
36	3	3 1/4	3 3/4	4	4 1/4	4 1/4	4 1/4

FLAPS 10, 15, 25

CENTER OF GRAVITY - PERCENT MAC	TAKEOFF GROSS WEIGHT - 1000 KG.						
	40	50	60	70	80	82	84
	STABILIZER TRIM UNITS						
6	6 3/4	8	8 1/2	8 1/2	8 1/2	8 1/2	8 1/2
8	6 1/2	7 1/2	8 1/2	8 1/2	8 1/2	8 1/2	8 1/2
10	6	7 1/4	8	8 1/2	8 1/2	8 1/2	8 1/2
12	5 3/4	6 3/4	7 1/2	8	8 1/4	8 1/4	8 1/4
14	5 1/4	6 1/4	7	7 1/2	7 3/4	7 3/4	7 3/4
16	5	6	6 1/2	7	7	7 1/4	7 1/4
18	4 1/2	5 1/2	6 1/4	6 1/2	6 3/4	6 3/4	6 3/4
20	4 1/4	5	5 3/4	6	6 1/4	6 1/2	6 1/2
22	3 3/4	4 3/4	5 1/4	5 3/4	6	6	6
24	3 1/4	4 1/4	4 3/4	5 1/4	5 1/2	5 1/2	5 1/2
26	3	3 3/4	4 1/2	4 3/4	5	5 1/4	5 1/4
28	3	3 1/2	4	4 1/4	4 3/4	4 3/4	4 3/4
30	3	3	3 1/2	4	4 1/4	4 1/4	4 1/4
32	3	3	3	3 1/2	3 3/4	3 3/4	3 3/4
34	3	3	3	3	3 1/2	3 1/2	3 1/2
36	3	3	3	3	3	3	3

AFMSTA801_REV_M_A300404BL3_D.RJENS_11-19-19

* DERATE RESULTING FROM ANY COMBINATION OF SELECTED PUSHBUTTON DERATE AND/OR ASSUMED TEMPERATURE THRUST REDUCTION

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

Regulatory Compliance A163806

The information in the basic Airplane Flight Manual and supplemented by this Appendix and in the Airplane Flight Manual - Digital Performance Information (AFM-DPI) identified by the Performance Software Serial Number(s) in the Log of Pages is presented in compliance with the requirements of EASA CS- 25.

Generation of EASA-approved performance information may be accomplished by use of the EASA-approved AFM-DPI software application. Any modification to the EASA-approved AFM-DPI software application and/or subsequent alterations to the generated output will cancel the approval of the information, unless this change was approved by the appropriate airworthiness authority. This will be applicable independently of the printed approval status on the generated output.

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

Revision Highlights

Revision 12 to D631A002-EASA dated 06/15/2021

The purpose of this revision is to remove the TCAS language from the 'Operational Data' chapter as it is revised and updated in the basic manual.

Also, this revision corrects the acronym 'CS' from Certification Standards to Certificate Specifications.

The following changes comprise this revision:

Chapter - Front Matter

Revision Approval (Revised)

This section uniquely identifies the approval authority and the reference number for this revision.

Log of Sections (Revised)

Revised to reflect sections changed in this revision.

Introduction (Revised)

Revised to correct the acronym 'CS' from Certification Standards to Certificate Specifications.

Chapter - Certificate Limitations

Kind of Airplane Operation (Revised)

Revised to correct the acronym 'CS' from Certification Standards to Certificate Specifications.

Landing on Wet Skid-Resistant Runways (Revised)

Revised to correct Certification Standards to Certification Specifications.

Chapter - Operational Data

Traffic Alert and Collision Avoidance System (TCAS) (Deleted)

This section is being deleted as the content is covered in the basic FAA certified manual.

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

Page Intentionally Left Blank

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

Title **A180846**

APPENDIX EASA

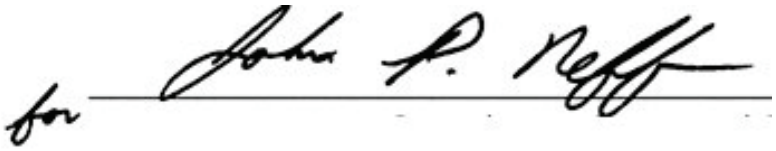
EUROPEAN AVIATION SAFETY AGENCY

MODEL 737-8/-8200/-9 SERIES

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

Revision Approval A211680

This Revision Approval may only be used in conjunction with a Log of Sections that refers to Revision Approval Number A211680 in the revision approval number column for the Revision Approval section.

Approved by:  Date:06/15/2021

Manager, Northwest Flight Test Section, AIR-715
Federal Aviation Administration
Seattle, WA

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

Log of Sections A211680

D631A002-EASA Revision 12 Date: 06/15/2021

	Chapter/Section	Revision Approval Number	Approval Date
	Front Matter		
	Title	A180846	03/25/2021
*	Revision Approval	A211680	06/15/2021
*	Log of Sections	A211680	06/15/2021
	Preface	A163357	03/23/2017
*	Introduction	A211680	06/15/2021
	Certificate Limitations		
	Certification Status	A163357	03/23/2017
	Flaps 15 Landing	A163357	03/23/2017
	Fuel System - Usable Fuel	A163357	03/23/2017
*	Kind of Airplane Operation	A211680	06/15/2021
*	Landing on Wet Skid-Resistant Runways	A211680	06/15/2021
	Required Navigation Performance (RNP)	A201358	12/21/2020
	Towing Operation	A180846	03/25/2021
	Operating Procedures		
	General	A211697	02/26/2021
	Airspeed Unreliable (E)	A201358	12/21/2020
	ALT Disagree	A211697	02/26/2021
	Engine - High Vibration	A170337	10/01/2018
	Engine Cowl Anti-Ice	A163357	03/23/2017
	Engine Cowl Valve Open - TAI Indication	A163357	03/23/2017
	Inoperative Airspeed Indicators	A170337	10/01/2018
	Loss of Both Engine Driven Generators	A163357	03/23/2017
	Main or Service Door Open in Flight	A163357	03/23/2017
	One Engine Inoperative Landing	A163357	03/23/2017
	Operation on Standby Electrical Power	A163357	03/23/2017
	Partial or Gear Up Landing	A163357	03/23/2017
	Probe Heat	A163357	03/23/2017
	Radio Altimeter Failure	A163357	03/23/2017
	Volcanic Cloud Avoidance (N)	A170202	06/05/2017
	Wing Thermal Anti-Ice Failure	A170337	10/01/2018
	Operational Data		
	Reduced Vertical Separation Minimum (RVSM)	A163357	03/23/2017
*	Traffic Alert and Collision Avoidance System (TCAS) - Deleted.	A170337	10/01/2018
	Performance		
	Definitions - Airspeeds	A163357	03/23/2017

* = Revised Sections

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

Log of Sections (Continued) A211680

Chapter/Section	Revision Approval Number	Approval Date
Definitions - Takeoff Path	A163357	03/23/2017
Flight with Unreliable Airspeed	A180846	03/25/2021
Minimum Wait Time After Landing	A170092	06/07/2017
Performance Conditions and Procedures - Landing Field Length	A163357	03/23/2017
Performance Conditions and Procedures - Takeoff	A180376	10/19/2020
Standard Performance Conditions	A170337	10/01/2018

* = Revised Sections

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

Preface A163357

This Airplane Flight Manual (AFM) appendix complies with the European Aviation Safety Agency (EASA) Certification Specifications (CS). This AFM appendix adds, amends, replaces, and/or supersedes information found in the U.S. Federal Aviation Administration (FAA) approved AFM.

This AFM appendix is approved by the EASA and signed by the FAA on behalf of the EASA.

Each revision to this AFM appendix includes a new Log of Sections listing the effective sections comprising this AFM appendix. Revisions to text material are indicated by vertical bars in the margins.

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

Introduction A211680

This appendix provides the additional certificate limitations, procedures and performance to operate the Boeing Model 737 MAX series airplane in accordance with the European Aviation Safety Agency (EASA) Certification Specifications (CS).

This appendix is used in conjunction with the Certificate Limitations, Operating Procedures, Operational Data and Performance of the basic Airplane Flight Manual and other appropriate appendices. The performance data is determined from the Airplane Flight Manual-Digital Performance Information (AFM-DPI) software identified by the Performance Software Serial Number(s) in the Performance Data appendix.

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

Certification Status A163357

This airplane is certificated in the Transport Category in accordance with EASA Certification Specifications (CS).

Flaps 15 Landing A163357

Flaps 15 normal landings are prohibited.

Fuel System - Usable Fuel A163357

Fuel remaining in the fuel tanks when quantity reads zero is not usable in flight.

Kind of Airplane Operation A211680

The airplane is approved for the following kinds of flight and operation, both day and night, when the required equipment is installed and approved in accordance with the applicable Certification Specifications (CS).

Visual (VFR)
Instrument (IFR)
Icing Conditions
Extended Over-Water*

*Operation over water at a horizontal distance of more than 50 nautical miles from the nearest shoreline.

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

Landing on Wet Skid-Resistant Runways A211680

The following is applicable to the landing performance in AFM-DPI provided as Alternate Performance for Wet Skid-resistant (WSR) Runways.

Flaps 30 and 40 wet landing distances for a wet porous friction course (PFC) and/or grooved runway are provided for the 737-8/9 model airplane. Porous friction course and grooved runway surfaces are designed to provide, when wet, a higher friction coefficient than conventional runways. The WSR landing performance provided was evaluated per EASA certification specifications and is only approved for use on runways having a grooved or PFC surface that has been constructed and maintained to acceptable standards. FAA AC 150/5320-12B “Measurement, Construction, and Maintenance of Skid Resistant Airport Paving Surfaces” provides guidance on such standards.

The inclusion of this data in the AFM does not constitute operational approval to base the landing performance requirements on these distances. Where such operational approval has not been obtained, the performance information applicable to a smooth, hard-surfaced runway must be used.

The data is not applicable to operations with Antiskid Inoperative.

Notes:
Specific airplane serial numbers must be operated in accordance with the Authorized Alternate Performance Capability specified in the basic Airplane Flight Manual.

Required Navigation Performance (RNP) A201358

Required Navigation Performance - Authorization Required (RNP AR)
RNP AR operations are prohibited.

Towing Operation A180846

Towbarless towing is prohibited unless the towbarless towing operations are performed in compliance with appropriate operational requirements. Towbarless towing must be conducted using towbarless tow vehicles that are designed and operated to preclude damage to the airplane nose wheel steering system, or that provide a reliable and unmistakable warning when damage to the steering system may have occurred. Only towbarless towing vehicles listed in Table 4 of Boeing Service Letter, 737-SL-09-002, Towbarless Towing, are specifically accepted for towbarless towing of the 737-8/-8200/9.

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

General A211697

The operating procedures contained in this manual have been developed and recommended by the manufacturer and approved by the EASA for use in the operation of this aircraft. These procedures are provided as guidance and should not be construed as prohibiting the development of equivalent approved procedures.

Procedures in this chapter are peculiar to the operation of the Model 737 airplane and are presented in abbreviated form. These procedures as well as additional procedures and information are found in the 737 Flight Crew Operating Manual for the operation of this aircraft.

This chapter contains "non-normal" procedures which arise out of a system malfunction or failure, and/or which involve the use of special systems or the alternative use of normal systems to protect crew and passengers from harm, and to maintain the airworthiness of the airplane. Emergency procedures requiring immediate crew awareness are designated by (E).

If the "HGS 6000" HUD is installed, the AOA Disagree alert will not show on the HUD. In an unreliable airspeed condition, use all available indications to determine a reliable airspeed. The AOA Disagree alert only shows on the PFD.

Flight crews should follow company specific Standard Operating Procedures (SOPs) for the handling of Non-Normal Situations. Company SOPs for handling Non-Normal situations must include at least the items mentioned below:

As a general overview of how non-normal situations should be conducted, consider maximum use of autoflight system to reduce workload, if available and appropriate. The flight crew should recognize and announce the non-normal situation to be acknowledged by other pilot.

Maintain airplane control: The Pilot Flying (PF) is to maintain airplane control and ensure that the flight path is under control. The Pilot Monitoring (PM) is to monitor the flight path.

Analyze the situation: The flight crew shall review all warning lights, caution lights, and other alert lights to identify the non-normal situation. Prioritize alerts and recommend course of action. The Pilot Flying (PF) acknowledges and confirms the Pilot Monitoring’s (PM’s) identification or recommendation.

Take the proper action: Do the NNC memory items based on each crewmember’s area of responsibility. Call for and complete the appropriate NNC. Review all warning lights, caution lights, and other alerts; complete other NNCs as required.

Evaluate the need to land: Assess status of the airplane. Review options for diversion or continued flight.

This chapter also contains "normal procedure", designated by (N), where an action or sequence of actions or prohibition of incorrect actions, which if not observed, could result in an adverse effect on the airworthiness of the airplane or the safety of the passengers and crew.

Procedures contained herein assume:

(a) When able, aural warnings are silenced.

Continued on next page

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

Section General, continued from previous page A211697

(b) Oxygen masks and smoke goggles (if installed) are donned when cabin altitude is excessive or ambient air is contaminated, and normal oxygen is selected if ambient air is not contaminated.

(c) Landing at nearest suitable airport is accomplished in the event of fire or persistent smoke.

(d) All flight crew members go on 100% oxygen whenever a hand-held fire extinguisher is to be discharged in the flight deck area, or when required because of smoke accumulation.

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

Airspeed Unreliable (E) A201358

Airspeed or Mach indications are suspected to be unreliable:

Recall:
If autopilot is engaged, disengage. If autothrottle is engaged, disengage. Set both F/D switches to off.
Set the following gear up pitch attitude and thrust:
Flaps extended: 10° and 80% N1
Flaps up: 4° and 75% N1

Reference:
PROBE HEAT switches check on.
The following indications are reliable: attitude, N1, ground speed, and radio altitude.
Notes:
1. Stick shaker, overspeed warning and airspeed low alerts may sound erroneously or simultaneously.
2. The flight path vector and pitch limit indicator may be unreliable on the PFD and HUD (as installed).
3. If the AOA indicator option is installed, the stick shaker indicator may be unreliable. AOA digital readout, analog needle, and approach reference band may be unreliable if the airspeed unreliable condition is caused by erroneous AOA.

Attempt to determine a reliable airspeed indication.

If a reliable airspeed indication can be determined:

Use the reliable airspeed indication for the remainder of the flight. If only the standby airspeed indication is reliable do not use autopilot, autothrottle, or flight directors. If the captain's or first officer's airspeed indication is reliable, turn on the flight director switch on the reliable side. If needed, engage autopilot on the reliable side. Do not use autothrottle.
Note: Autopilot may not engage or may disengage automatically.

If a reliable airspeed indication cannot initially be determined:

Using performance tables from an approved source, set the pitch attitude and thrust setting for the current airplane configuration and phase of flight. When in trim and stabilized, compare the captain, first officer, and standby airspeed indicators with the airspeed shown in the table. An airspeed indication that differs by more than 20 knots or 0.03 Mach from the airspeed shown in the table should be considered unreliable. If only the standby airspeed indication is reliable, do not use autopilot, autothrottle, or flight directors. If the captain's or first officer's airspeed indication is reliable, turn on the flight director switch on the reliable side, and autopilot if needed. Do not use autothrottle.
Note: Autopilot may not engage or may disengage automatically.

If a reliable airspeed indication cannot be determined:

Using the performance tables from an approved source, set pitch attitude and thrust setting for the airplane configuration and phase of flight as needed. Reference an approved source for landing distances.
Notes:
1. Maintain visual conditions if possible.
2. Establish landing configuration early.
3. Radio altitude reference is available below 2500 feet.

Continued on next page

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

Section Airspeed Unreliable (E), continued from previous page A201358

4. Use electronic and visual glideslope indicators, where available, for approach and landing.

Attempt to determine a reliable altitude indication.

Use the most reliable altitude indication for the remainder of the flight. If the captain's or first officer's altitude indication is reliable:
The airplane may not meet RVSM requirements. Set transponder to reliable side and select traffic alerts only mode.

If captain's and first officer's altitude indications are both unreliable:
Turn off transponder altitude reporting.
Note: Airplane does not meet RVSM requirements.

A nuisance stick shaker may be deactivated at pilot's discretion. This improves recognition of a stall warning on the opposite side.
Note: Elevator Feel Shift may be active, resulting in increased control column forces.

If deactivating stick shaker **is** needed: Only the active stick shaker should be deactivated.
Deactivate nuissance stick shaker.
Notes:
1. When the circuit breaker is pulled, increased control column forces due to Elevator Feel Shift activation are removed.
2. The stick shaker on the opposite side is not deactivated.

If deactivating stick shaker **is not** needed; end of procedure except deferred items.

In addition to the normal descent, approach and landing checklists, complete the following deferred items:

For approach, only set the BARO minimums on the reliable PFD. Remove the BARO minimums from the unreliable PFD.
Note: If BARO minimums are set only on the First Officer's PFD, DA/MDA aural callouts are not provided. Use the performance tables from an approved source to determine the go-around pitch attitude and thrust setting.

In the event of a go-around if either the Captain's or First Officer's airspeed indication is reliable, when TO/GA is pushed, the flight director pitch bar may automatically be removed. An AFDS pitch mode change, such as LVL CHG, restores the flight director pitch bar.
Note: Only use flight director guidance on the reliable PFD.

In the event of a go-around and the standby airspeed indication is the only reliable airspeed, do not use TO/GA.

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

ALT Disagree A211697

The ALT DISAGREE alert is displayed on the captain's and first officer's altitude tape on the PFD when the indications disagree.
If the IAS DISAGREE alert is also shown on the speed tape of the PFD, accomplish the Airspeed Unreliable procedure.
If the IAS DISAGREE is not shown, check all altimeters are set to correct barometric setting.
If the ALT DISAGREE alert extinguishes, continue normal operation.
If the ALT DISAGREE alert remains, standby altimeter is available, do not use the flight path vector, maintain visual conditions if possible and if a reliable altitude is determined, use the transponder for the reliable side.
Note: Airplane does not meet RVSM requirements.
If a reliable altitude is not determined, set the transponder to not transmit altitude.

In addition to the normal descent, approach and landing checklists, complete the following deferred items:

For approach, only set the BARO minimums on the reliable PFD. Remove the BARO minimums from the unreliable PFD.
Note: If BARO minimums are set only on the First Officer's PFD, DA/MDA aural callouts are not provided.
Establish landing configuration early.
Radio altitude reference is available below 2,500 ft.
Use electronic and visual glideslope indicators where available for approach and landing.

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

Engine - High Vibration A170337

If VIB levels are 4.0 units or greater accompanied by airframe vibrations, accomplish the following:

If not in icing conditions:

Disengage autothrottle and reduce thrust on affected engine to maintain VIB below 4.0 units.

Note: If the VIB indication does not decrease when the thrust lever is retarded, check other engine indications on the affected engine. If other engine indications are normal, run the engine at reduced thrust.

If in icing conditions:

Ground Operations in Icing Conditions:

A. During ground operations (including taxi in, taxi out, and ground hold) in icing conditions (OAT +3 °C or less) or if increased fan vibration due to fan ice accumulation is present, the flight crew must accomplish the following ice shed procedures:

1. Increase thrust to a minimum of 50% N1.
2. Decrease thrust to idle and repeat as necessary or at intervals no greater than 60 minutes.

Note: Engine vibration may indicate above the normal operating range up to the maximum display value during ice shedding, however, this will have no adverse effect on the engine.

B. Within 5 minutes of or in conjunction with the takeoff:

1. Increase thrust to a minimum of 50% N1.
2. Maintain that thrust level for 5 seconds minimum.
3. Confirm stable engine operation prior to the final advance to takeoff thrust.

Note: Engine vibration may indicate above the normal operating range up to the maximum display value during ice shedding, however, this will have no adverse effect on the engine.

In Flight Icing Conditions:

1. Reduce thrust on one engine at a time to 45% N1. After approximately 5 seconds, advance thrust lever to a minimum of 80% N1 while monitoring engine vibrations.
2. Reduce thrust as needed for flight conditions.
3. Wait 15 seconds. This allows engine vibration level to stabilize.
4. If vibration does not decrease accomplish the procedure for "if not in icing conditions".
5. If above steps successfully remove fan ice and vibration is less than 4.0 units after thrust is reduced, repeat at approximately 15 minutes intervals or as needed.
6. If vibration decreases, indicating ice removal, resume normal operation.

Note: Engine vibration may indicate above the normal operating range up to the maximum display value during ice shedding, however, this will have no adverse effect on the engine.

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

Engine Cowl Anti-Ice A163357

An engine COWL ANTI-ICE light illuminated indicates an overpressure condition in the cowl anti-ice duct. Reduce thrust until the COWL ANTI-ICE light extinguishes (flight conditions permitting) and disengage autothrottle (if engaged).

Engine Cowl Valve Open - TAI Indication A163357

An engine COWL VALVE light remaining illuminated amber in flight and/or an amber TAI indication the cowl anti-ice valve position disagrees with the engine anti-ice switch position.

Valve Open - If total air temperature is above 10 °C, limit thrust on the affected engine to 80% N1 if possible.

Valve Closed - Avoid icing conditions.

Inoperative Airspeed Indicators A170337

Erratic or inconsistent airspeed indications should not cause any control difficulties. Fly the airplane using attitude, rather than airspeed, as the primary reference. Normal flight and approach landing can be accomplished using attitude, rather than airspeed, in conjunction with normal thrust and airplane configuration. Avoid causing rapid changes in speed and airplane configuration.

Note: Refer to Flight with Unreliable Airspeed Section in the Performance Chapter of this Appendix.

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

Loss of Both Engine Driven Generators A163357

All TRANSFER BUS OFF, SOURCE OFF, and GEN OFF BUS lights illuminated indicate the loss of both engine driven generators.

Note: With main tank fuel pumps inoperative above 30,000 feet, thrust deterioration or engine flameout may occur.

Engine Generator Switches	ON

If only one SOURCE OFF light extinguishes:

APU (if available)	START & ON BUS

If both SOURCE OFF lights remain illuminated:

Bus Transfer Switch	OFF
Electric Hydraulic Pump Switches	OFF
APU (if available)	START & ON BUSSES
Bus Transfer Switch	AUTO

If both primary attitude displays are inoperative:

IRS Mode Selector Switches	ATT

Maintain wings level, constant speed flight until attitude displays recover (approximately 30 seconds).

Magnetic Heading	ENTER

If both SOURCE OFF lights remain illuminated:

Avoid icing conditions.

Note: Flight in icing conditions may result in erroneous flight instrument indications.

Land at the nearest suitable airport.

Note : Fully charged batteries provide a minimum of 60 minutes of standby power.

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

Main or Service Door Open in Flight A163357

A door annunciator light illuminated indicates an exterior door is not properly latched. Check door handle of affected door and position to close if necessary. If the door handle is in the closed position and cabin pressurization is normal, proceed normally. If the door handle will not close and/or cabin pressurization is not normal, land at nearest suitable airport.

One Engine Inoperative Landing A163357

Use flaps 15 and V_{REF} 15 for landing approach, land using flaps 15. If any of the following conditions apply, set $V_{REF\ ICE} = V_{REF}$ 15 + 10:

Engine anti-ice will be used during landing.
Wing anti-ice has been used any time during the flight.
Icing conditions were encountered during the flight and the landing temperature is below 10°C.

Maintain V_{REF} 15 + 5 or $V_{REF\ ICE}$ + 5 minimum on final approach to assure adequate maneuver margin and speed for go-around. Do not add more than 10 knots wind correction to $V_{REF\ ICE}$.

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

Operation on Standby Electrical Power A163357

When operating on standby power only, a minimum of 60 minutes of electrical power is available from the two batteries. DC power is supplied to the hot battery bus, switched hot battery bus, the battery bus and the standby DC bus. 115-volt AC power is supplied from the static inverter to the standby AC bus and the 28-volt AC standby bus is supplied from a transformer off the 115-volt AC standby bus.

Services provided for include: Flight instruments, including Captain's attitude/heading system, airspeed, Mach, barometric altitude, vertical speed, #1 navigation and communication radios, #1 ADF, #1 stall warning, standby altimeter/airspeed, and standby attitude with ILS/GLS* deviation bars. The Captain's outboard display unit, left display electronics unit, EFIS control panel are active. As an optional feature one or more of the following additional items may be provided for; #1 ATC, #1 DME, GPS, #1 FMC/CDU and Captain's inboard display unit.

*__Note__: GLS is an optional feature and may or may not be installed.

The Integrated Standby Flight Display (ISFD) System includes a dedicated battery that provides an additional 150 minutes minimum of ISFD operation following the loss of standby power (main battery depletion).

APU and engine generator controls and indicators.

Electronic engine control, engine and APU starting, ignition, fuel shutoff and crossfeed valves and fuel quantity indicators, engine N1 and EGT indicators, primary display on the center upper display unit, and hydraulic fluid shutoff valves.

Takeoff warning, landing gear indication and warning, and antiskid INOP lights.

Master fire warning, engine and APU fire detection, warning and extinguishing systems.

Thrust reversers, inboard antiskid system and parking brake light.

Oxygen indicators and valve.

Air-conditioning overheat indicators, pack valves and DC manual pressurization control.

Standby flood and compass lights, master caution and integral lights for standby flight instruments, digital clock, control cabin white dome lights, and forward air stair standby controls and actuator.

Passenger address, and flight and service interphone.

Pitot probes, the total air temperature probe and the alpha vanes are electrically heated. Static ports are not heated. When operating on standby power, only the captain's pitot probe is heated, however, the CAPT PITOT light does not illuminate for a failure.

Emergency area lights, floor proximity lights and exit signs, which are powered by battery packs independent of the main battery.

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

Partial or Gear Up Landing A163357

If all landing gear do not indicate down and locked after attempting manual gear extension, accomplish the following procedures. Brief crew and passengers on emergency landing and evacuation procedures. Burn off fuel to reduce touchdown speed. Plan a flaps 40 landing. Set V_{REF} 40.

Descent - Approach

Anti-Ice	AS REQUIRED
Air Conditioning & Pressurization	SET
Altimeters & Instruments	SET & X-CHECKED
N1 & IAS BUGS	CHECKED & SET, V_{REF} 40

Prior to Approach

Aural Warn C/B (P6-3)	PULL
Auto Speed Brake C/B (P6-2)	PULL
Ground Proximity Gear Inhibit Switch	GEAR INHIBIT
Auto Brake Select Switch	OFF
Available Gear	EXTEND IF DESIRED
Engine Bleed Air Switches	OFF
APU Switch	OFF
Landing Procedure	REVIEW
Position fuel pump switches OFF just prior to flare. After stop, evacuate passengers.	

Note: Do not raise the speed brakes unless stopping distance is critical.

Landing

Engine Start Switches	ON
Recall	CHECKED
Speed Brake	DOWN DETENT
Landing Gear	ALL AVAILABLE DOWN GREEN LIGHT(S)
Flaps	40, GREEN LIGHT

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

Probe Heat A163357

Illumination of the probe heat light(s) indicates related probe is not heated. Avoid icing conditions. Flight in icing condition may result in erroneous flight instrument indications.

Radio Altimeter Failure A163357

Category II or III approach and autoland are prohibited.

Volcanic Cloud Avoidance (N) A170202

Avoid operations in potential volcanic cloud environments. Operation in a volcanic cloud, even for short time, can cause significant aircraft malfunctions and damage to the aircraft.

Wing Thermal Anti-Ice Failure A170337

In flight, if either or both valves fail in the closed position in icing conditions, exit the icing conditions. This is indicated by either or both L VALVE or R VALVE lights continuously illuminating amber with the Wing Anti-Ice switch ON. Turn the switch OFF and avoid icing conditions.

For approach and landing Flaps 30 or 40 with a WTAI valve failed in the closed position: If icing conditions are encountered during any part of the flight and the forecast landing temperature is below 10 °C, or if ice formations are observed on the airplane surfaces, (wings, windshield wipers, window frames, etc.) set final approach (landing) speed to V_{REF} 30 + 10 knots or V_{REF} 40 + 10 knots. The combined airspeed corrections for wind, gust, and icing should not exceed a maximum of 15 knots.

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

Reduced Vertical Separation Minimum (RVSM) A163357

The type design reliability and performance of this airplane/engine combination has been evaluated in accordance with EASA Information Leaflet No. 23 dated April 1994, “Interim Guidance Material on the Approval of Operators/Aircraft for RVSM Operators” and FAA Memorandum 91-RVSM, dated 14 March 1994, "Interim Guidance for Approval of Aircraft for Reduced Vertical Separation Minima (RVSM) Flight" and is qualified for operation in RVSM airspace. This finding does not constitute approval to conduct reduced vertical separation minimum operations.

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

Page Intentionally Left Blank

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

Definitions - Airspeeds A163357

All airspeeds and Mach values in AFM-DPI are based on the primary airspeed system and assume zero instrument error.

Static source position error, ΔV_p, is the error due to the location of the static port within the airplane pressure field.

Indicated Airspeed, V_I, IAS - Airspeed reading, as displayed in the airplane.

Calibrated Airspeed, V_C, CAS - Indicated airspeed corrected for static source position error ($V_C = V_I + \Delta V_p$).

Equivalent Airspeed, V_e, EAS - Calibrated airspeed corrected for compressibility ($V_e = V_C - \Delta V_C$).

True Airspeed, V_T, TAS - Equivalent airspeed corrected for atmospheric density effects ($V_T = V_e/\sqrt{\sigma}$).

True Mach Number, M - Machmeter reading, as installed in the airplane, corrected for static source position error

Critical Engine Failure Speed, V_{EF} - The speed at which, if the most critical engine fails, the engine failure is recognized at V_1, one second later.

Takeoff Decision Speed, V_1 - The speed from which a decision to continue the takeoff, following a failure of the critical engine at V_{EF}, results in a takeoff distance to a height of 35 feet at V_2 speed that will not exceed the usable takeoff distance and the speed at which the first action is taken to bring the airplane to a full stop that will not exceed the accelerate-stop distance available. V_1 must not be less than the minimum V_1 for control on the ground, $V_{1\,(MCG)}$, or greater than the rotation speed, V_R, or greater than the maximum brake energy speed, V_{MBE}.

Rotation Speed, V_R - The speed at which rotation from the three-point attitude to the takeoff attitude is initiated. The scheduled rotation speed must be equal to or greater than 1.05 V_{MCA} and V_1.

Maximum Brake Energy Speed, V_{MBE} - The highest takeoff decision speed from which the airplane may be brought to a stop without exceeding the maximum energy absorption capability of the brakes. Maximum Brake Energy Speed is compared in takeoff planning to V_1 speed.

Takeoff Safety Speed, V_2 - The target speed to be attained at the 35 foot height, assuming recognition of an engine failure at or after V_1 . This speed must be at least 1.13 times the reference stall speed in the takeoff configuration.

V_{MCA} - The minimum flight speed at which the airplane is controllable with a maximum of 5 degrees bank when the critical engine suddenly becomes inoperative with the remaining engine at takeoff thrust.

Continued on next page

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

Section Definitions - Airspeeds, continued from previous page A163357

Minimum V_1 for Control on the Ground, $V_{1\,(MCG)}$ - The minimum takeoff decision speed, V_1, at which, when the critical engine suddenly becomes inoperative at V_{EF} with the remaining engine at takeoff thrust, it is possible to control the airplane with primary aerodynamic controls alone and continue the takeoff.

Minimum Control Speed for Landing, V_{MCL} - The minimum control speed with the airplane configured for approach at which the airplane is controllable with a maximum of 5 degrees bank when the critical engine suddenly becomes inoperative with the remaining engine at takeoff thrust.

Landing Speed - The minimum speed at the 50 foot height in a normal landing. This speed must be at least 1.23 times the reference stall speed in the landing configuration and not less than V_{MCL}.

Reference Landing Speed, V_{REF} - This speed is equal to the landing speed in the flaps 40 landing configuration. This speed must not be less than V_{MCL}.

Design Maneuvering Speed, V_A - The speed above which full application of rudder, ailerons or elevator, or maneuvers involving angles of attack near stall, should be avoided.

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

Definitions - Takeoff Path A163357

The takeoff path assumes failure of the most critical engine at V_{EF} and extends from a standing start to a point where the airplane is at least 1500 feet above the takeoff surface and has achieved the enroute configuration and final climb speed.

The takeoff path is divided into segments representing distinct changes in airplane configuration, airspeed, and engine thrust as defined below. Some of these definitions are abbreviated versions of those appearing in EASA Certification Specifications (CS); the appropriate regulations should be consulted for complete details.

Clearway - An obstacle-free area beyond the takeoff runway which can be used as a part of the takeoff distance available.

Stopway - An area beyond the takeoff runway capable of supporting the airplane in an aborted takeoff which can be used as part of the accelerate-stop distance available. To be applicable for use with this manual, the surface characteristics of the stopway must not differ substantially from those of the runway.

Takeoff Distance Required - The greater of:

1. The distance to takeoff and climb to a height of 35 feet (dry runway[1]) with a failure of the critical engine at V_{EF};

or

2. 115 percent of the distance to takeoff and climb to a height of 35 feet with all engines operating.

Takeoff Run Required - The greater of:

1. The distance to takeoff and climb to a point equidistant between lift off and the 35 foot height point (dry runway[2]) with a failure of the critical engine at V_{EF};

or

2. 115 percent of the distance to takeoff and climb to a point equidistant between lift off and the 35 foot height point with all engines operating.

Accelerate-Stop Distance Required - The greater of:

1. The sum of the distances required to accelerate with all engines operating and come to a complete stop assuming a critical engine failure at V_{EF}

or

2. The sum of the distances required to accelerate with all engines operating and come to a complete stop with all engines still operating.

Notes:

1. On a wet runway, the height requirement with a failed engine is 15 feet.
2. On a wet runway, the takeoff run required is the distance to takeoff and climb to 15 feet with a failure of the critical engine at V_{EF}.

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.

Flight with Unreliable Airspeed A180846

For model 737-8/-8200: Altitude and/or vertical speed indications may also be unreliable.

Climb (280/.76)
Flaps Up, Set Max Climb Thrust

PRESSURE ALTITUDE (FT)		WEIGHT (1000 KG)					
		40	50	60	70	80	90
40000	PITCH ATT V/S (FT/MIN)	**4.0** 1700	**4.0** 1100	**4.5** 600			
30000	PITCH ATT V/S (FT/MIN)	**4.0** 2500	**4.0** 1900	**4.0** 1500	**4.0** 1100	**4.0** 800	**4.0** 600
20000	PITCH ATT V/S (FT/MIN)	**7.0** 4000	**6.5** 3100	**6.0** 2500	**6.0** 2100	**6.0** 1700	**6.0** 1300
10000	PITCH ATT V/S (FT/MIN)	**11.0** 5500	**9.5** 4300	**8.5** 3500	**8.0** 2900	**7.5** 2400	**7.5** 2100
SEA LEVEL	PITCH ATT V/S (FT/MIN)	**15.0** 6800	**12.5** 5300	**11.0** 4300	**10.0** 3600	**9.5** 3100	**9.5** 2600

Cruise (.76/280)
Flaps Up, Set Thrust for Level Flight

PRESSURE ALTITUDE (FT)		WEIGHT (1000 KG)					
		40	50	60	70	80	90
40000	PITCH ATT %N1	**2.0** 78.4	**2.5** 80.9	**3.5** 84.5			
35000	PITCH ATT %N1	**1.5** 76.8	**2.0** 78.2	**2.5** 80.0	**3.0** 82.5	**3.5** 85.7	
30000	PITCH ATT %N1	**1.0** 76.2	**1.5** 77.1	**2.0** 78.3	**2.5** 79.8	**3.0** 81.6	**3.5** 84.2
25000	PITCH ATT %N1	**1.0** 72.2	**1.5** 73.0	**2.0** 74.3	**2.5** 75.7	**3.0** 77.6	**4.0** 79.8
20000	PITCH ATT %N1	**1.0** 68.1	**1.5** 69.0	**2.0** 70.1	**3.0** 71.5	**3.5** 73.2	**4.0** 75.4
15000	PITCH ATT %N1	**1.0** 64.3	**1.5** 65.0	**2.5** 66.1	**3.0** 67.3	**3.5** 68.9	**4.0** 70.9

Descent (.76/280)
Flaps Up, Set Idle Thrust

PRESSURE ALTITUDE (FT)		WEIGHT (1000 KG)					
		40	50	60	70	80	90
40000	PITCH ATT V/S (FT/MIN)	**-1.0** -2400	**0.0** -2200	**0.5** -2200	**1.5** -2300	**1.5** -2800	**1.0** -3600
30000	PITCH ATT V/S (FT/MIN)	**-3.0** -3000	**-2.0** -2500	**-1.0** -2200	**0.0** -2000	**0.5** -1900	**1.0** -1900
20000	PITCH ATT V/S (FT/MIN)	**-3.5** -2900	**-2.0** -2400	**-1.0** -2100	**0.0** -1900	**0.5** -1800	**1.5** -1700
10000	PITCH ATT V/S (FT/MIN)	**-3.5** -2700	**-2.0** -2200	**-1.0** -1900	**0.0** -1800	**0.5** -1600	**1.5** -1600
SEA LEVEL	PITCH ATT V/S (FT/MIN)	**-4.0** -2400	**-2.5** 2000	**-1.5** -1800	**-0.5** -1600	**0.5** -1500	**1.0** -1400

Continued on next page

Boeing Proprietary. Copyright © The Boeing Company. May be subject to export restrictions under EAR. See title section for details.